Abstract

This work proposes a refined lattice Boltzmann scheme for coupled thermal-flow problems governed by the weakly compressible Navier-Stokes equations and the convection-diffusion equation, with a rigorous and consistent treatment up to the second order in Mach number. Source terms are introduced and designed by an inverse design strategy, in order to provide tunable parameters to the lattice Boltzmann equations (LBE). This approach is similar to the concept of the lattice kinetic scheme (LKS); however, the latter modifies the equilibrium distribution function. The derivation in the present work begins from the Boltzmann equation, providing a solid physical foundation for this LKS-like operator, which has rarely been addressed in related problems. By adjusting the parameters, the relaxation time in the lattice Boltzmann method (LBM) can be shifted away from 1/2 under conditions of low viscosity and diffusivity, thereby enhancing the numerical stability of the Bhatnagar-Gross-Krook model. Through the Chapman-Enskog expansion, the present scheme can fully recover the macroscopic governing equations. The numerical results demonstrate the validity and the superior accuracy relative to previous LKS-like methods, together with its stability improvement over the standard LBM, as evidenced by simulation results for several thermal flow problems. It is noteworthy that careful selection of the tunable parameters is essential in balancing model accuracy and numerical stability and the guidelines for choosing these parameters are proposed accordingly. Clearly, this work takes advantage of the higher phase-space dimensionality of LBE to expand the application of LBM to thermal flows at high Reynolds numbers and Péclet numbers.

Keywords: Lattice Boltzmann method, Inverse design, Source term design, Numerical stability, Heat and mass transfer

1. Introduction

The combination of the Navier-Stokes (N-S) equations and the convection-diffusion (C-D) equation is widely employed to solve hydrodynamic flows and heat/mass transfer problems. The strongly non-linear characteristics of both N-S equations and C-D equation, together with the increasing computational resources in recent decades, lead to the proliferation of numerical simulation tools, such as the finite difference method, the finite volume method, and the finite element method, for solving these coupled equations [1–3]. The lattice Boltzmann method (LBM) is one of the alternative and promising approaches based on the kinetic theory [4]. Unlike traditional numerical methods that directly solve the macroscopic equations, LBM is a mesoscopic approach solving the Boltzmann equation, where macroscopic variables are obtained by the integral moments of the particle distribution function [5]. In other words, LBM simulates the macroscopic hydrodynamic and scalar equations in a higher-dimensional phase space, offering unique algorithmic design flexibility that allows the method to handle a wide variety of flow problems with essentially the same formulation. Using source terms to introduce design flexibility, as shown in this work, is such an example. Furthermore, the Cartesian-grid layout and the lattice structure of LBM simplify the coding and are particularly well suited to GPU acceleration [6]. Several studies report substantial performance gains, exceeding one order of magnitude speedup compared with CPU implementations, resulting in markedly reduced computational time and improved fault tolerance [7–9]. These unique advantages of LBM drive its rapid development, and the effective integration with other techniques leads to widespread applications in engineering problems [10–15].

LBM was initially proposed by McNamara and Zanetti in 1988, originating from the lattice gas automata but overcoming its statistical noise problem [16, 17]. By incorporating the Bhatnagar-Gross-Krook (BGK) collision operator [18], the single-relaxation-term LBM has been sequentially but independently proposed by several research groups [19–21]. This development based on the simple linearized collision model, facilitates the simulation efficiency and flexibility. Later, He and Luo [22, 23] proved that LBM can be rigorously derived from the continuous Boltzmann equation, establishing a solid foundation for LBM.

The convection-diffusion problem, which integrates the N-S equations and C-D equation, is one of the major applications of LBM simulations. Several early attempts to develop LBM schemes for this problem have been documented [24–26]. Among various approaches, the double-distribution-function (DDF) approach has garnered significant attention owing to its excellent performance in numerical stability and its ability to adjust Prandtl (Pr) / Schmidt (Sc) numbers compared to the multispeed approach [5, 27–31]. However, despite the widespread use of the BGK collision operator, numerical instability at low viscosity or diffusivity remains a concern. This instability is closely linked to the value of the relaxation time in LBM; specifically, low viscosity or diffusivity causes the relaxation time to approach 0.5, resulting in significant numerical instability.

Substantial efforts have been devoted to addressing this instability problem. Inamuro [32] was the first to propose the lattice kinetic scheme (LKS) to tackle the instability problem in incompressible thermal flow. By incorporating additional terms related to the shear rate and temperature gradient into the equilibrium distribution functions, the relaxation time can be fixed at unity, and the proposed scheme possesses superior numerical stability compared to the standard LBM. This idea has been sequentially applied to non-Newtonian flow [33] and two-phase flow [34]. Later, Wang *et al.* [35] proposed a modified LKS scheme by adding a correction term of the form $-A\Delta t\boldsymbol{S} : c_s^2\boldsymbol{I}$ in the equilibrium distribution, overcoming the lack-of-mass-conservation issue that arises in a non-Newtonian flow when the trace of the shear rate is non-zero. So far, this refined LKS with this additional correction in equilibrium distribution has been extended to simulate multiphase flow [36], porous media flow [37], and convection-diffusion problem [38]. For the convection-diffusion problem, Yang *et al.* [39] incorporated the correction term into the source term of the lattice Boltzmann equation (LBE), demonstrating a good accuracy and capability in dealing with high Rayleigh numbers, but they ignored the impact of the source term in the C-D equation and formulated their work based on the low Mach number (Ma) assumption. The scheme proposed by Wang *et al.* [38] involved a scalar source and demonstrated the enhancement of the improved LKS scheme in terms of numerical stability. However, although their schemes can mathematically recover the macroscopic governing equations to some extent, they introduced additional terms without rigorous physical derivations. Thus far, to the best of the authors' knowledge, the LBE incorporating a well-designed source term to solve the convection-diffusion problem at low viscosity or diffusivity has not yet been fully analyzed in terms of achieving better numerical stability and minimal truncation errors.

To address this research gap and facilitate the broader application of LBM, a refined LBM model for the convection-diffusion problem is proposed through inverse design [40, 41], with the LKS-like terms acting as the source terms in LBE. Inverse design denotes a process opposite to conventional derivations, by first specifying the target macroscopic governing equations, which corresponds to the moments of the model Boltzmann equation, for both flow and scalar fields in this work. Using Chapman–Enskog analysis and enforcing the moment constraints, we can obtain the source terms that fully recover N-S equations and C-D equation, thereby providing a systematic account of the rationale and derivation of the resulting formulations, rigorously truncated at second order in the Mach number. The obtained source terms were then implemented in the LBE framework and validated against benchmark tests. To improve the computational efficiency, the simulations in this work are accelerated using NVIDIA Tesla P100 GPUs.

The remainder of the paper is organized as follows: The inverse design of the source terms in the continuous Boltzmann equation is detailed in Sec. 2, along with the subsequent expressions of the source terms in the LBE and the formulations of the shear rate and scalar gradient. The proposed scheme is validated through several benchmark cases and its effectiveness in improving numerical stability is demonstrated in Sec. 3. Finally, the main conclusions are summarized in Sec. 4.

2. Methodology

In general, the design of LBM consists of two parts: (1) the proper formulation of the LBE for the interior nodes, (2) a consistent formulation of the implementation of the macroscopic boundary conditions. The correct formulation of the source term in the Boltzmann equation for both the flow and scalar fields can be derived by the inverse design matching the moment equations of the Boltzmann equation with the N-S equations and C-D equation. Sec. 2.1 presents the macroscopic equations, and Sec. 2.2 details the systematic derivation of the source terms in the Boltzmann equation through inverse design. Subsequently, Sec. 2.3 shows the expressions for source terms in the LBE, and Sec. 2.4 provides the formulations for shear rate and scalar gradient in terms of the non-equilibrium distribution functions. Finally, Sec. 2.5 formulates the boundary conditions employed in this work.

2.1. Macroscopic hydrodynamic equations

The macroscopic equations for the typical convection-diffusion problems, including the weakly compressible N-S equations for the flow field and C-D equation for the scalar field, are given by

$$\partial_t \delta\rho + \partial_\alpha (\rho_0 u_\alpha) = 0, \tag{1a}$$

$$\partial_t (\rho_0 u_\alpha) + \partial_\beta \left(\rho_0 u_\alpha u_\beta\right) = -\partial_\alpha p + \partial_\beta \sigma_{\alpha\beta} + F_\alpha, \tag{1b}$$

$$\partial_t T + \partial_\alpha (T u_\alpha) = \partial_\alpha (\kappa \partial_\alpha T) + Q, \tag{1c}$$

where ρ_0 is the mean fluid density and $\delta\rho$ is the density fluctuation. u, p, and T specify the flow velocity, pressure, and the variable in the scalar field (could be the temperature or concentration), respectively. κ represents the diffusivity in the scalar field. F_α and Q denote the external body force and heat/mass source, respectively. $\sigma_{\alpha\beta}$ denotes the viscous stress tensor, which is expressed as

$$\sigma_{\alpha\beta} = 2\mu \left(S_{\alpha\beta} - \frac{1}{D} S_{\gamma\gamma}\delta_{\alpha\beta}\right) + \mu^B S_{\gamma\gamma}\delta_{\alpha\beta}, \tag{2}$$

where D, μ, and μ^B represent the dimensions of the physical problem, the shear viscosity, and the bulk viscosity, respectively. $S_{\alpha\beta} = \frac{1}{2}\left(\partial_\alpha u_\beta + \partial_\beta u_\alpha\right)$ defines the strain rate tensor, and $S_{\gamma\gamma} = \partial_\gamma u_\gamma$ is the velocity divergence.

2.2. Generalized formulation of the source terms by inverse design

2.2.1. The generalized source-term design for the flow field

The continuous Boltzmann equation for flow field with the BGK collision operator and a source term $\mathcal{S}$ can be written as [18, 22, 23]

$$\partial_t f + v_\alpha \partial_\alpha f = -\frac{1}{\tau_f}(f - f^{eq}) + \mathcal{S}, \tag{3}$$

where f (i.e., $f(\boldsymbol{v}, \boldsymbol{x}, t)$) denotes the distribution function of the particle velocity $\boldsymbol{v}$ at location $\boldsymbol{x}$ and time t. τ_f represents the relaxation time, associated with shear viscosity. The equilibrium distribution function f^{eq} in this work is based on the scheme proposed by He and Luo [42], utilizing a truncated Hermite expansion, as

$$f^{eq} = \omega \left\{\delta\rho + \rho_0 \left[\frac{v_\alpha u_\alpha}{c_s^2} + \frac{u_\alpha u_\beta \left(v_\alpha v_\beta - c_s^2 \delta_{\alpha\beta}\right)}{2c_s^4}\right]\right\}, \tag{4}$$

where c_s is the sound speed. The weighting function is $\omega \equiv (2\pi c_s^2)^{-D/2} \exp(-v^2/2c_s^2)$, which satisfies the following conditions

$$\int \omega d\boldsymbol{v} = 1, \int v_\alpha v_\beta \omega d\boldsymbol{v} = c_s^2 \delta_{\alpha\beta}, \int v_\alpha v_\beta v_\gamma v_\mu \omega d\boldsymbol{v} = c_s^4 \left(\delta_{\alpha\beta}\delta_{\gamma\mu} + \delta_{\alpha\gamma}\delta_{\beta\mu} + \delta_{\alpha\mu}\delta_{\gamma\beta}\right). \tag{5}$$

Note that all odd-order moments of ω are zero by symmetry. Therefore, the first four moments of the equilibrium distribution function f^{eq} are given by

$$\int f^{eq} d\boldsymbol{v} = \delta\rho, \; \int v_\alpha f^{eq} d\boldsymbol{v} = \rho_0 u_\alpha, \; \int v_\alpha v_\beta f^{eq} d\boldsymbol{v} = \delta\rho c_s^2 \delta_{\alpha\beta} + \rho_0 u_\alpha u_\beta,$$
$$\int v_\alpha v_\beta v_\gamma f^{eq} d\boldsymbol{v} = \rho_0 c_s^2 \left(u_\alpha \delta_{\beta\gamma} + u_\beta \delta_{\alpha\gamma} + u_\gamma \delta_{\alpha\beta} \right). \tag{6}$$

The above formulation preserves the moments of the original Maxwellian equilibrium, with a leading-order truncation error at $O(Ma^3)$. Next, the concept of inverse design is employed to formulate the source term $\mathcal{S}$ properly. The idea is to require the zeroth and the first-order moments of Eq. (3) to reproduce, respectively, the continuity equation and the N-S equations, leading to the following requirements

$$\partial_t \delta\rho + \partial_\alpha (\rho_0 u_\alpha) = 0, \; \rightarrow \int S \, d\boldsymbol{v} = 0, \tag{7a}$$

$$\partial_t (\rho_0 u_\alpha) + \partial_\beta \int v_\alpha v_\beta f \, d\boldsymbol{v} = F_\alpha, \; \rightarrow \int v_\alpha S \, d\boldsymbol{v} = F_\alpha, \tag{7b}$$

in which,

$$\begin{aligned} \partial_\beta \int v_\alpha v_\beta f d\boldsymbol{v} &= \partial_\beta \int v_\alpha v_\beta f^{eq} d\boldsymbol{v} + \partial_\beta \int v_\alpha v_\beta (f - f^{eq}) \, d\boldsymbol{v} \\ &= \partial_\beta \left(\delta\rho c_s^2 \delta_{\alpha\beta} \right) + \partial_\beta \left(\rho_0 u_\alpha u_\beta \right) + \partial_\beta \int v_\alpha v_\beta (f - f^{eq}) \, d\boldsymbol{v}. \end{aligned} \tag{8}$$

By employing the Chapman-Enskog expansion, the distribution function f can be approximated by

$$f = f^{eq} - \tau_f \left(\partial_t f^{eq} + v_\alpha \partial_\alpha f^{eq} - \mathcal{S} \right) + O\left(\tau_f^2 \right), \tag{9}$$

and by comparing with Eq. (1b), the viscous stress tensor $\sigma_{\alpha\beta}$ is calculated as

$$\begin{aligned} \sigma_{\alpha\beta} &= -\int v_\alpha v_\beta (f - f^{eq}) \, d\boldsymbol{v} = \tau_f \int v_\alpha v_\beta \left(\partial_t f^{eq} + v_\gamma \partial_\gamma f^{eq} - \mathcal{S} \right) d\boldsymbol{v} \\ &= \tau_f u_\alpha F_\beta + \tau_f u_\beta F_\alpha - \tau_f c_s^2 u_\alpha \partial_\beta \delta\rho - \tau_f c_s^2 u_\beta \partial_\alpha \delta\rho - \tau_f \rho_0 u_\alpha u_\gamma \partial_\gamma u_\beta - \tau_f \rho_0 u_\beta u_\gamma \partial_\gamma u_\alpha \\ &\quad -2\tau_f \rho_0 u_\alpha u_\beta \partial_\gamma u_\gamma + \tau_f \rho_0 c_s^2 \partial_\beta u_\alpha + \tau_f \rho_0 c_s^2 \partial_\alpha u_\beta - \tau_f \int v_\alpha v_\beta \mathcal{S} d\boldsymbol{v}, \end{aligned} \tag{10}$$

where the continuity equation and the Euler equations have been used to remove the time derivative terms. Matching the target constitutive equation (Eq. (2)), the second-order moment of the source term ($\mathcal{S}$) should be designed to yield

$$\begin{aligned}
\int v_\alpha v_\beta \mathcal{S}\, d\boldsymbol{v} = 2\left(\rho_0 c_s^2 - \frac{\mu}{\tau_f}\right) S_{\alpha\beta} \qquad & O\left(Ma^1\right) \\
+ \frac{1}{\tau_f}\left(\frac{2}{D}\mu - \mu^B\right) S_{\gamma\gamma}\delta_{\alpha\beta} \qquad & O\left(Ma^2\right) \\
+ u_\alpha F_\beta + u_\beta F_\alpha \qquad & O\left(Ma^2\right) \\
- c_s^2 u_\alpha \partial_\beta \delta\rho - c_s^2 u_\beta \partial_\alpha \delta\rho \qquad & O\left(Ma^3\right) \\
- \rho_0 u_\alpha u_\gamma \partial_\gamma u_\beta - \rho_0 u_\beta u_\gamma \partial_\gamma u_\alpha \qquad & O\left(Ma^3\right) \\
- 2\rho_0 u_\alpha u_\beta \partial_\gamma u_\gamma. \qquad & O\left(Ma^4\right)
\end{aligned} \tag{11}$$

Ignoring the terms at the third or higher order in Mach number, Eq. (11) provides the second-order moment for $\mathcal{S}$. With the zeroth (Eq. (7a)), first (Eq. (7b)), and second-order moments together, the source term $\mathcal{S}$ can be formulated by the truncated Hermite expansion up to the second order, as

$$\begin{aligned}
\mathcal{S} = \mathcal{S}^{H2} = \omega &\left[\frac{F_\alpha v_\alpha}{c_s^2}\left(1 + \frac{v_\beta u_\beta}{c_s^2}\right) - \frac{F_\beta u_\beta}{c_s^2}\right] \\
&+ \omega \frac{v_\alpha v_\beta - c_s^2 \delta_{\alpha\beta}}{c_s^4}\left[\left(\rho_0 c_s^2 - \frac{\mu}{\tau_f}\right) S_{\alpha\beta} + \frac{1}{\tau_f}\left(\frac{1}{D}\mu - \mu^B\right) S_{\gamma\gamma}\delta_{\alpha\beta}\right].
\end{aligned} \tag{12}$$

Each term in this formulation can be properly interpreted. The first term represents the corrected formulation of the external forcing, as discussed in Guo *et al.* [43]. The second term is used to adjust the shear viscosity, whose default value is $\mu^{\text{default}} = p\tau_f \approx p_0\tau_f = \rho_0 c_s^2 \tau_f$. The third term is introduced here to adjust the bulk viscosity, whose default value is $\mu^{B,\text{default}} = 2\mu^{\text{default}}/D = 2\tau_f \rho_0 c_s^2/D$, a value usually non-zero in kinetic models. The way the shear viscosity is adjusted here through the source term resembles the lattice kinetic scheme, with the difference that this correction is done in LKS by modifying the equilibrium distribution.

More specifically, we can re-write Eq. (12) as

$$\mathcal{S} = \omega\left[\frac{F_\alpha v_\alpha}{c_s^2}\left(1 + \frac{v_\beta u_\beta}{c_s^2}\right) - \frac{F_\beta u_\beta}{c_s^2}\right] + \rho_0 \omega \left(\frac{v_\alpha v_\beta}{c_s^2} - \delta_{\alpha\beta}\right)\left[\chi_s S_{\alpha\beta} + \frac{1}{D}(\chi_b - \chi_s) S_{\gamma\gamma}\delta_{\alpha\beta}\right], \tag{13}$$

where the two dimensionless free parameters, χ_s and χ_b, are introduced to adjust the resulting shear and bulk viscosity independently. Specifically, the kinematic shear and bulk viscosity are given by

$$\nu \equiv \frac{\mu}{\rho_0} \equiv (1 - \chi_s)\nu^{\text{default}} = (1 - \chi_s)\tau_f c_s^2, \quad \nu^B \equiv \frac{\mu^B}{\rho_0} \equiv (1 - \chi_b)\nu^{\text{B,default}} = (1 - \chi_b)\frac{2\tau_f c_s^2}{D}. \tag{14}$$

Obviously, the case of $\chi_s = \chi_b = 0$ corresponds to the default model. Since the Second Law of Thermodynamics implies that the two viscosities cannot be negative, it requires $\chi_s \leq 1$ and $\chi_b \leq 1$. The relaxation time τ_f should be chosen to ensure numerical stability of the LBE. In this work, χ_b is set to zero in all the cases. On the other hand, if we would like to follow the Stokes assumption that $\nu^B = 0$, then we could set $\chi_b = 1$.

2.2.2. *The generalized source-term design for the scalar field*

Similarly, the continuous Boltzmann equation for the scalar field, such as temperature or concentration, incorporating the BGK collision operator and the source term $\mathcal{R}$, is formulated as [18, 44]

$$\partial_t g + v_\alpha \partial_\alpha g = -\frac{1}{\tau_g}\left(g - g^{eq}\right) + \mathcal{R}, \tag{15}$$

where g (i.e., $g(\boldsymbol{v}, \boldsymbol{x}, t)$) and g^{eq} are the distribution function and its corresponding equilibrium distribution function for the scalar field. τ_g is a separate relaxation time. The expression of g^{eq} adopted here is proposed by Chai and Zhao [3], which can effectively achieve the derivative of the scalar field from the local non-equilibrium distribution function, thereby eliminating any truncation errors caused by the non-local finite-differencing operation [32, 45]. This expression of g^{eq} is written as

$$g^{eq} = \omega T\left[1 + \frac{v_\alpha u_\alpha}{c_s^2} + \frac{u_\alpha u_\beta\left(v_\alpha v_\beta - c_s^2\delta_{\alpha\beta}\right)}{2c_s^4}\right] + \lambda\frac{Tp}{\rho_0 c_s^2} \tag{16}$$

where λ is a new weighting function, introduced to provide a non-zero contribution only to the second-order moment, namely, $\int \lambda d\boldsymbol{v} = 0$, $\int \lambda v_\alpha d\boldsymbol{v} = \boldsymbol{0}$, $\int \lambda v_\alpha v_\beta d\boldsymbol{v} = c_s^2\delta_{\alpha\beta}$. This is best done through the weighting factors for the discrete velocities, by modifying the weighting factor of the zero velocity, relative to the corresponding discrete weighting factors associated with ω. Consequently, the first three moments of g^{eq} can be expressed as

$$\int g^{eq} d\boldsymbol{v} = T, \quad \int v_\alpha g^{eq} d\boldsymbol{v} = T u_\alpha, \quad \int v_\alpha v_\beta g^{eq} d\boldsymbol{v} = T c_s^2\delta_{\alpha\beta} + T u_\alpha u_\beta + \frac{Tp}{\rho_0}\delta_{\alpha\beta}. \tag{17}$$

Now, the same inverse-design method is employed to determine the source term $\mathcal{R}$. Namely, the zeroth-order moment of Eq. (15) is used to match the C-D equation, leading to the following requirement

$$\partial_t T + \partial_\alpha \int v_\alpha g d\boldsymbol{v} = Q, \rightarrow \int \mathcal{R} d\boldsymbol{v} = Q, \tag{18}$$

in which,

$$\int v_\alpha g d\boldsymbol{v} = \int v_\alpha g^{eq} d\boldsymbol{v} + \int v_\alpha\left(g - g^{eq}\right) d\boldsymbol{v} = T u_\alpha + \int v_\alpha\left(g - g^{eq}\right) d\boldsymbol{v}. \tag{19}$$

The distribution function g can be approximated using the leading-order Chapman-Enskog expansion

$$g = g^{eq} - \tau_g\left(\partial_t g^{eq} + v_\alpha \partial_\alpha g^{eq} - \mathcal{R}\right) + O\left(\tau_g^2\right). \tag{20}$$

Matching with Eq. (1c), the flux term $\kappa\partial_\alpha T$ is obtained as

$$\begin{aligned}\kappa\partial_\alpha T &= -\int v_\alpha\left(g - g^{eq}\right) d\boldsymbol{v} = \tau_g \int v_\alpha\left(\partial_t g^{eq} + v_\beta\partial_\beta g^{eq} - \mathcal{R}\right) d\boldsymbol{v} \\ &= \tau_g T\frac{F_\alpha}{\rho_0} + \tau_g p\frac{\partial_\alpha T}{\rho_0} + \tau_g c_s^2\partial_\alpha T + \tau_g u_\alpha Q - \tau_g T u_\alpha \partial_\beta u_\beta - \tau_g \int v_\alpha R d\boldsymbol{v}.\end{aligned} \tag{21}$$

Therefore, the first-order moment of $\mathcal{R}$ is given by

$$\begin{aligned}\int v_\alpha R d\boldsymbol{v} = & \left(\frac{p}{\rho_0} + c_s^2 - \frac{\kappa}{\tau_g}\right)\partial_\alpha T & O\left(Ma^0\right) \\ & + T\frac{F_\alpha}{\rho_0} + u_\alpha Q & O\left(Ma^1\right) \\ & - T u_\alpha \partial_\beta u_\beta. & O\left(Ma^3\right)\end{aligned} \tag{22}$$

Ignoring the term at the order $O\left(Ma^3\right)$, Eq. (22) provides the first-order moment of $\mathcal{R}$. Then, the source term $\mathcal{R}$ can be formulated based on the zeroth-order (Eq. (18)) and first-order moments, through the first-order truncated Hermite expansion, as

$$\mathcal{R} = \mathcal{R}^{H1} = \omega\left\{\frac{v_\alpha\left(p\partial_\alpha T + TF_\alpha\right)}{\rho_0 c_s^2} + Q\left(1 + \frac{v_\alpha u_\alpha}{c_s^2}\right) + \frac{v_\alpha}{c_s^2}\left(c_s^2 - \frac{\kappa}{\tau_g}\right)\partial_\alpha T\right\}. \tag{23}$$

This source-term design includes three parts. The first part represents the necessary coupling term with the N-S equations, identical to the formulation of Chai and Zhao [3]. The second part is used to accurately model the heat source term Q in the C-D equation, Eq. (1c). The third part provides the ability to adjust the scalar diffusivity from its default value $\kappa^{\text{default}} = \tau_g c_s^2$, which in spirit is similar to the LKS treatment for adjustable shear viscosity. Eq. (23) clearly extends the formulation of Chai and Zhao [3], and conceptually aligns with the work of Chen *et al.* [40] who formulated the source terms for a fully compressible flow. Here we treat only the limit of weakly compressible flow with the scalar C-D equation.

Eq. (23) can be re-written as

$$\mathcal{R} = \omega\left\{\frac{v_\alpha\left(p\partial_\alpha T + TF_\alpha\right)}{\rho_0 c_s^2} + Q\left(1 + \frac{v_\alpha u_\alpha}{c_s^2}\right) + v_\alpha \chi_\kappa \partial_\alpha T\right\}, \tag{24}$$

in which a free parameter χ_κ has been introduced. Then the model scalar diffusivity is

$$\kappa \equiv \left(1 - \chi_\kappa\right)\kappa^{\text{defult}} = \left(1 - \chi_\kappa\right)\tau_g c_s^2. \tag{25}$$

Again, the case of $\chi_\kappa = 0$ represents the standard, unaltered model, and, in general, all χ_κ values less than one are allowed.

2.3. Source terms in the Lattice Boltzmann equation

Subsequently, the source terms $\mathcal{S}$ and $\mathcal{R}$ in the Boltzmann equations for the flow and scalar fields are transformed into their corresponding expressions in the LBEs, Φ and Ψ, respectively, thereby facilitating implementations. The derivation of Φ and Ψ from $\mathcal{S}$ and $\mathcal{R}$ follows Eqs. (A.4) and (A.16), with detailed rationales given in Appendix A.

Therefore, the LBE source term Φ obtained from the expression in Eq. (13) is written as

$$\begin{aligned}\Phi &= \left(1-\frac{1}{2\tau_{fL}}\right)\mathcal{S}\\ &= \omega\left(1-\frac{1}{2\tau_{fL}}\right)\left[\frac{F_\alpha v_\alpha}{c_s^2}\left(1+\frac{v_\beta u_\beta}{c_s^2}\right)-\frac{F_\beta u_\beta}{c_s^2}\right]\\ &+\rho_0\omega\left(1-\frac{1}{2\tau_{fL}}\right)\left(\frac{v_\alpha v_\beta}{c_s^2}-\delta_{\alpha\beta}\right)\left[\chi_s S_{\alpha\beta}+\frac{1}{D}\left(\chi_b-\chi_s\right)S_{\gamma\gamma}\delta_{\alpha\beta}\right],\end{aligned} \tag{26}$$

where $\tau_{fL} = \tau_f/dt + 0.5$. The first term is fully consistent with the forcing term proposed by Guo *et al.* [43], but the derivation is made much simpler by starting with the continuous Boltzmann equation instead of the LBE.

Without the additional term $\chi_s S_{\alpha\beta}$ in Eq. (26), the shear viscosity is given by $\mu = \mu^{\text{default}} = \rho_0\nu = \tau_f\rho_0 c_s^2$ in the standard LBE, which inevitably leads to numerical instability when the kinematic viscosity ν approaches 0, implying $\tau_{fL} \to 0.5$ [46].

For our generalized model introduced in the previous section, the viscosity is given by

$$\nu = (1-\chi_s)\tau_f c_s^2 = (\tau_{fL}-0.5)(1-\chi_s)c_s^2 dt,\;\; \nu^B = 2(1-\chi_b)\tau_f c_s^2/D = 2(\tau_{fL}-0.5)(1-\chi_b)c_s^2 dt/D \tag{27}$$

For small viscosity values, it is not necessary to have τ_{fL} close to 0.5, since χ_s can now be used to achieve a small viscosity even we fix τ_{fL} to one as in LKS. Using a larger τ_{fL} is known to enhance the numerical stability of LBM [32, 38].

The source term Ψ for the scalar LBE is derived from the formulation in Eq. (24) and transferred according to the relation in Eq. (A.16), which is given by

$$\begin{aligned}\Psi &= \left(1-\frac{1}{2\tau_{gL}}\right)\mathcal{R}\\ &= \omega\left(1-\frac{1}{2\tau_{gL}}\right)\left\{\frac{v_\alpha\left(p\partial_\alpha T+TF_\alpha\right)}{\rho_0 c_s^2}+Q\left(1+\frac{v_\alpha u_\alpha}{c_s^2}\right)+v_\alpha\chi_\kappa\partial_\alpha T\right\},\end{aligned} \tag{28}$$

where $\tau_{gL} = \tau_g/dt + 0.5$.

Similar to the flow field, if the default expression $\kappa = c_s^2\tau_g$ is adopted, the value of τ_{gL} approaches 0.5 as κ approaches 0, leading to the numerical instability. In our generalized model, however, the diffusivity is given as

$$\kappa = (1-\chi_\kappa)\,\tau_g c_s^2 = (\tau_{gL}-0.5)\,(1-\chi_\kappa)\,c_s^2 dt. \tag{29}$$

Once again, we can achieve a small diffusivity by adjusting the parameter χ_κ while maintaining τ_{gL} away from 0.5. This approach can also enhance the numerical stability of the scalar LBE.

2.4. Computation of the shear rate and scalar gradient

The transformation from the Boltzmann distribution to the lattice Boltzmann distribution in the flow field, as expressed in Eq. (A.2), can be used to obtain the LBEs ((Eqs. (A.3) and (A.15), respectively) from the continuous Boltzmann-BGK equations (Eqs. (3) and (15)). The combination of this transformation (Eq. (A.2)) with the formulations for viscous stress tensor (Eqs. (2) and (10)) leads to the following relation between the second-order non-equilibrium moment and the strain rate

$$\int v_\alpha v_\beta \left(\widetilde{f} - f^{eq}\right) d\boldsymbol{v} = \left(1 + \frac{dt}{2\tau_f}\right) \int v_\alpha v_\beta \left(f - f^{eq}\right) d\boldsymbol{v} - \frac{dt}{2} \int v_\alpha v_\beta S \, d\boldsymbol{v}$$
$$= -2\mu S_{\alpha\beta} + \left(\frac{2}{D}\mu - \mu^B\right) S_{\gamma\gamma}\delta_{\alpha\beta} - \frac{dt}{2}\left[u_\alpha F_\beta + u_\beta F_\alpha + 2\rho_0 c_s^2 S_{\alpha\beta}\right], \tag{30}$$

where $\widetilde{f}$ is related to the transformed distribution solved in LBE. Involving the free parameters χ_s and χ_b in LBE (see Eq. (27)), the strain rate can be calculated from the non-equilibrium distribution, as

$$S_{\alpha\beta} = -\frac{2\int v_\alpha v_\beta \left(\widetilde{f} - f^{eq}\right) d\boldsymbol{v} + dt\left(u_\alpha F_\beta + u_\beta F_\alpha\right)}{4\mu + 2dt\rho_0 c_s^2}, \quad \text{for } \alpha \neq \beta. \tag{31}$$

To obtain S_{11}, S_{22}, S_{33}, we first obtain the velocity divergence $S_{\gamma\gamma}$ from Eq. (30) as

$$S_{\gamma\gamma} = -\frac{\int v_\gamma v_\gamma \left(\widetilde{f} - f^{eq}\right) d\boldsymbol{v} + dt u_\gamma F_\gamma}{D\mu^B + dt\rho_0 c_s^2}, \tag{32}$$

It follows then

$$S_{11} = -\frac{\int v_1 v_1 \left(\widetilde{f} - f^{eq}\right) d\boldsymbol{v} + dt u_1 F_1}{2\mu + dt\rho_0 c_s^2} - \frac{2\mu - D\mu^B}{D(2\mu + dt\rho_0 c_s^2)} S_{\gamma\gamma}. \tag{33}$$

S_{22} and S_{33} can be computed similarly. In conclusion, all components of the strain rate can be obtained from the non-equilibrium distribution in LBE.

Similarly, using the transformation in the scalar field (see Eq. (A.14)), the first-order moment of the non-equilibrium distribution function in LBE is given by

$$\int v_\alpha \left(\widetilde{g} - g^{eq}\right) d\boldsymbol{v} = -\left(1 + \frac{dt}{2\tau_g}\right)\kappa \partial_\alpha T - \frac{dt}{2}\left[\frac{p\partial_\alpha T + TF_\alpha}{\rho_0} + u_\alpha Q + \left(c_s^2 - \frac{\kappa}{\tau_g}\right)\partial_\alpha T\right]. \tag{34}$$

Then the gradient of the scalar field can be obtained by

$$\partial_\alpha T = -\frac{2\int v_\alpha \left(\widetilde{g} - g^{eq}\right) d\boldsymbol{v} + dt\left(TF_\alpha/\rho_0 + u_\alpha Q\right)}{2\kappa + dt(p/\rho_0 + c_s^2)} = -\frac{2\int v_\alpha \left(\widetilde{g} - g^{eq}\right) d\boldsymbol{v} + dt\left(TF_\alpha/\rho_0 + u_\alpha Q\right)}{c_s^2\left[2\tau_{gL}(1 - \chi_\kappa) + \chi_\kappa\right] dt + p/\rho_0 dt}. \tag{35}$$

As χ_κ approaches 0 and Q equals 0, Eq. (35) yields the same expression as Eq. (31) derived by Chai and Zhao [3].

2.5. Boundary conditions

As shown in [47], it is important to develop the mesoscopic wall boundary condition that is consistent with the internal LBE up to the second order. However, with the BGK collision operator, it is not possible to achieve this level of consistency [47]. Here, for both stationary and moving velocity wall boundaries, the standard halfway bounce-back scheme [48],

$$\widetilde{f}_i(\boldsymbol{x}_b, t+dt) = \widetilde{f}_{\bar{i}}^*(\boldsymbol{x}_b, t) + 2\omega_i \rho_0 \frac{v_{\alpha,i} u_{w,\alpha}}{c_s^2}, \tag{36}$$

is implemented, where $\widetilde{f}^*$ denotes the post-collision distribution function, while $\bar{i}$ represents the opposite direction of i, such that $v_{\alpha,\bar{i}} = -v_{\alpha,i}$. Additionally, $u_{w,\alpha}$ refers to the wall-boundary velocity. It is possible to improve the halfway bounce-back scheme using the two-relaxation-time collision model following the analysis of Dong *et al.* [47].

The anti-bounce-back scheme is employed at the isothermal boundary [49], as

$$\widetilde{g}_i(\boldsymbol{x}_b, t+dt) = -\widetilde{g}_{\bar{i}}^*(\boldsymbol{x}_b, t) + 2g_i^{eq}(\boldsymbol{x}_w, t+dt). \tag{37}$$

Similarly, the $\widetilde{g}^*$ denotes the post-collision distribution function in the scalar field. Considering the equilibrium distribution function in Eq. (16), for a stationary wall, Eq. (37) can be expressed as

$$\boxed{\widetilde{g}_i(\boldsymbol{x}_b, t+dt) = -\widetilde{g}_{\bar{i}}^*(\boldsymbol{x}_b, t) + 2\omega_i T + 2\lambda_i \frac{Tp}{\rho_0 c_s^2}.} \tag{38}$$

The halfway bounce-back scheme is also applied at adiabatic or impermeable boundaries [50, 51], as

$$\widetilde{g}_i(\boldsymbol{x}_b, t+dt) = \widetilde{g}_{\bar{i}}^*(\boldsymbol{x}_b, t). \tag{39}$$

3. Results and discussion

In this section, we shall first validate the formulation of source terms in the LBEs, Φ for the flow field and Ψ for the scalar field, by thermal flow problems such as thermal Poiseuille flow and thermal Couette flow with viscous dissipation. This paper employs the D2Q9 lattice velocity set for both the velocity field and the scalar field, for all the simulations in two dimensions. The D2Q9 velocity set achieves fifth-order Gauss-Hermite quadrature (GHQ) accuracy. Since the highest moment order for f^{eq} is third order, rigorously speaking a sixth-order GHQ accuracy is required. However, it is a common practice to use D2Q9 for the velocity field due to its simplicity as long as the numerical Mach number is kept small. On the other hand, for the scalar field, D2Q9 provides adequate GHQ accuracy since the highest moment order for g^{eq} is second order, and as such a fourth-order GHQ accuracy is sufficient. For the same reason, the D3Q27 lattice velocity set is used for the flow field, while three velocity sets (D3Q15, D3Q19, and D3Q27) are compared for the scalar field in three-dimensional (3D) simulations. The numerical accuracy in each case is assessed by comparing the converged numerical results with the analytical solution, using the relative error defined as

$$E_V = \sqrt{\frac{\sum_{\boldsymbol{x}} |V_a(\boldsymbol{x}) - V_n(\boldsymbol{x})|^2}{\sum_{\boldsymbol{x}} |V_a(\boldsymbol{x})|^2}}, \tag{40}$$

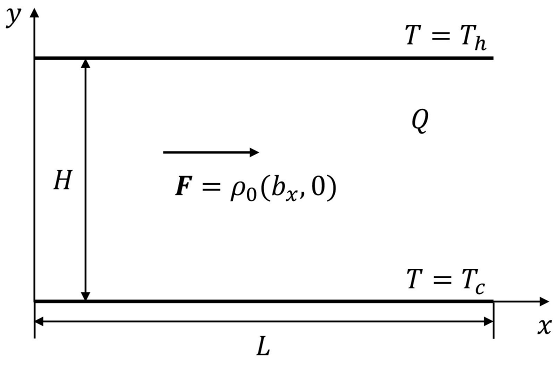

Figure 1: Schematic of the thermal Poiseuille flow

where V denotes the variable used for error evaluation, with the subscript "a" indicating the analytical solution and the subscript "n" indicating the numerical solution. For steady-flow cases, the criterion for terminating the simulation is defined as

$$\sqrt{\frac{\sum_x |\boldsymbol{u}(\boldsymbol{x}, t + 10000dt) - \boldsymbol{u}(\boldsymbol{x}, t)|^2}{\sum_x |\boldsymbol{u}(\boldsymbol{x})|^2}} \leq 10^{-6}, \tag{41a}$$

$$\frac{\sum_x |T(\boldsymbol{x}, t + 10000dt) - T(\boldsymbol{x}, t)|}{\sum_x |T(\boldsymbol{x})|} \leq 10^{-6}. \tag{41b}$$

Finally, the stability enhancement of the current improved model is discussed in the following part, using the natural convection problem as a benchmark.

3.1. Thermal Poiseuille flow

Thermal Poiseuille flow refers to a fully developed two-dimensional thermal channel flow with a height of H, driven by the constant body force $\boldsymbol{F} = \rho_0 (b_x, 0)$. As illustrated in Fig. 1, the upper and lower boundaries are stationary walls, with the upper boundary designated as the hot wall at a temperature of $T = T_h$, and the lower boundary as the cold wall at $T = T_c$. The heat source Q arises from viscous heat dissipation and is defined by the equation

$$Q = \frac{2\nu}{C_v} S_{\alpha\beta} S_{\alpha\beta}, \tag{42}$$

where C_v denotes the specific heat at constant volume. This scenario is governed by Eq. (1) and is characterized by three key dimensionless parameters: the Reynolds number Re, the Prandtl number Pr, and the Eckert number Ec. These three dimensionless numbers are defined as

$$Re = \frac{u_0 H}{\nu}, Pr = \frac{\nu}{\kappa}, Ec = \frac{u_0^2}{C_v (T_h - T_c)}, \tag{43}$$

where u_0 represents the maximum horizontal velocity and is associated with the body force, as

$$u_0 = \frac{b_x H^2}{8\nu}. \tag{44}$$

Using u_0 as the reference velocity and $\Delta T = T_h - T_c$ as the reference temperature, the analytical dimensionless velocity $(\bar{u}, \bar{v}) = (u, v)/u_0$ and temperature $\bar{T} = (T - T_c)/\Delta T$ for this problem at the steady state are respectively expressed as

$$\bar{u} = 4\bar{y}\left(1 - \bar{y}\right), \bar{v} = 0, \tag{45a}$$

$$\bar{T} = \bar{y} + \frac{1}{3} PrEc\left[1 - (1 - 2\bar{y})^4\right], \tag{45b}$$

where $\bar{y} = y/H$. Correspondingly, the velocity and temperature gradients are respectively expressed as

$$\frac{\partial \bar{u}}{\partial \bar{y}} = 4\left(1 - 2\bar{y}\right), \tag{46a}$$

$$\frac{\partial \bar{T}}{\partial \bar{y}} = \Delta T\left[1 + \frac{8}{3} PrEc\left(1 - 2\bar{y}\right)^3\right]. \tag{46b}$$

For the numerical solution, the computational domain is defined as $H/L = 1.0$. Various values for dimensionless parameters are considered, including $Re = 20$ and 100; $Pr = 0.71$; and $Ec = 10$, 50, and 100. The halfway bounce-back scheme is applied to the upper and lower fixed walls, while the anti-bounce-back scheme is utilized for the isothermal boundary. The periodic boundary condition is implemented on the left and right boundaries.

Figure 2 presents a comparison of the numerical profiles at steady state with the analytical solutions, including the dimensionless horizontal velocity $\bar{u}$, temperature $\bar{T}$, and their respective derivatives along the vertical midline $x/L = 0.5$ at $Pr = 0.71$ and $Re = 20$ for the default model $\chi_s = \chi_\kappa = 0$. Three distinct scenarios, with Ec values at 10, 50, and 100, are considered. A grid resolution of 100×100 is applied. $\partial\bar{u}/\partial\bar{y}$ and $\partial\bar{T}/\partial\bar{y}$ are locally calculated using the shear rate expression in Eq. (31) ($\partial\bar{u}/\partial\bar{y} = 2S_{12}$ as the vertical velocity $\bar{v}$ is zero) and the temperature gradient in Eq. (35). It is evident that the numerical results of $\bar{u}$, $\bar{T}$, $\partial\bar{u}/\partial\bar{y}$, and $\partial\bar{T}/\partial\bar{y}$ are all an excellent agreement with the respective analytical solution.

The local error of the numerical solution along the vertical midline $x/L = 0.5$ is presented in Fig. 3. Note that the error for the horizontal velocity $\bar{u}$ is independent of the Reynolds number Re and the Eckert number Ec, remaining constant along the vertical midline. Meanwhile, the error in gradient $\partial\bar{u}/\partial\bar{y}$ is negligible and close to the machine round-off error. Comparatively, the errors for temperature $\bar{T}$ and its gradient $\partial\bar{T}/\partial\bar{y}$ vary with the coordinate y but remain constant at the same value of Ec.

Subsequently, the spatial accuracy of the velocity and temperature is assessed. The analysis considers the following parameters: $Re = 20$ and 100; $Pr = 0.71$; and $Ec = 10$, 50, and 100. Grid resolutions of 50×50, 100×100, 150×150, and 200×200 are examined. The relative error is calculated using Eq. (40), where the variable V is substituted with $\bar{u}$ and $\bar{T}$. Fig. 4 presents the relative error for $\bar{u}$ and $\bar{T}$ versus the mesh dimension. The spatial error for $\bar{u}$ and $\bar{T}$ exhibits a standard slope of 2, indicating second-order accuracy for both velocity and temperature. Among them, the horizontal velocity error $E_{\bar{u}}$ remains constant as Re and Ec vary, consistent with results in Fig. 3(a). This observation precisely aligns with the theoretical prediction of the wall slip velocity for the halfway bounce-back scheme analyzed in Dong *et al.* [47].

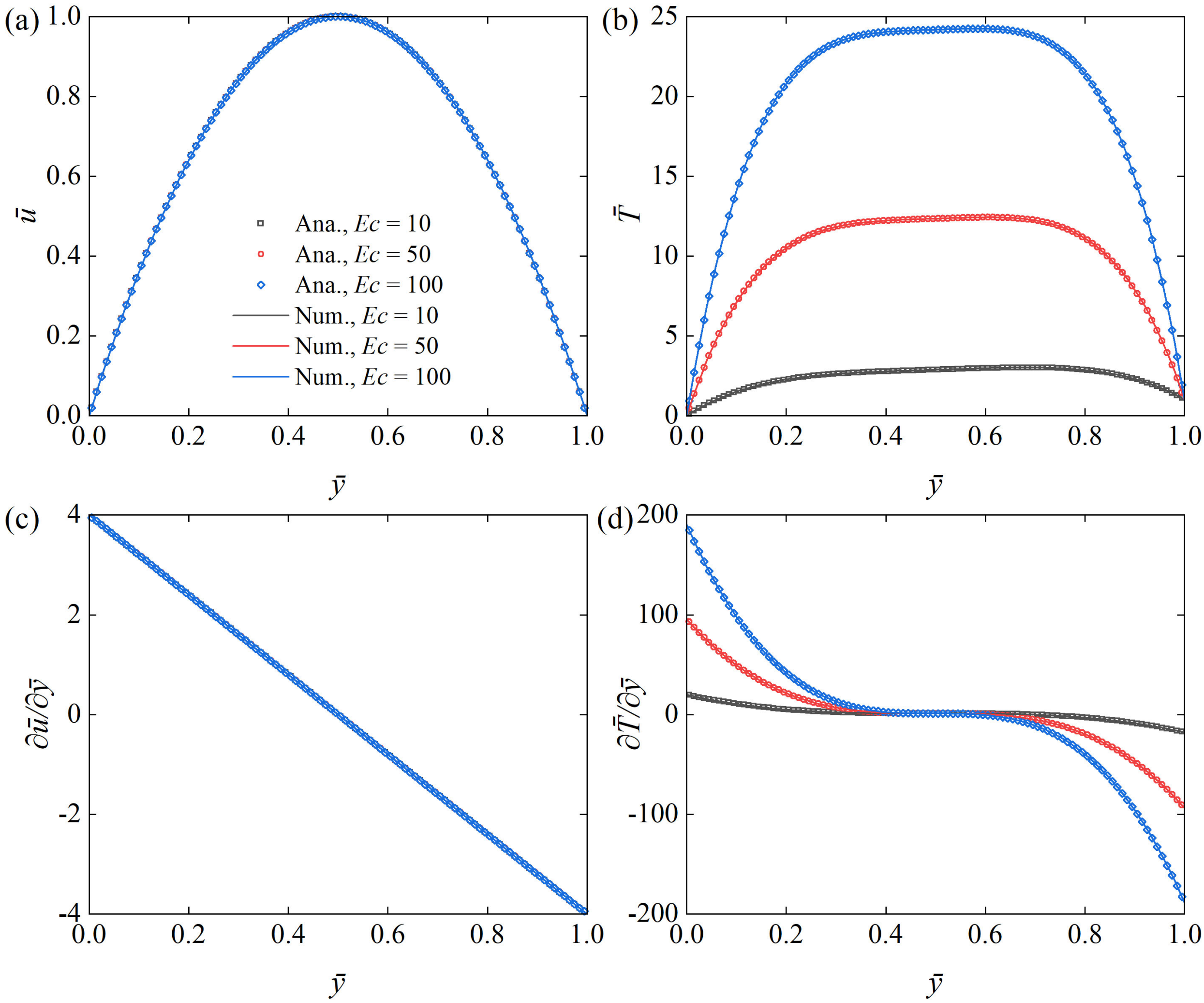

Figure 2: Comparison of various profiles between the analytical and numerical solutions for (a) the horizontal velocity $\bar{u}$, (b) the temperature $\bar{T}$, (c) the velocity gradient $\partial\bar{u}/\partial\bar{y}$, and (d) the temperature gradient $\partial\bar{T}/\partial\bar{y}$ at Pr = 0.71 and Re = 20 along the vertical midline.

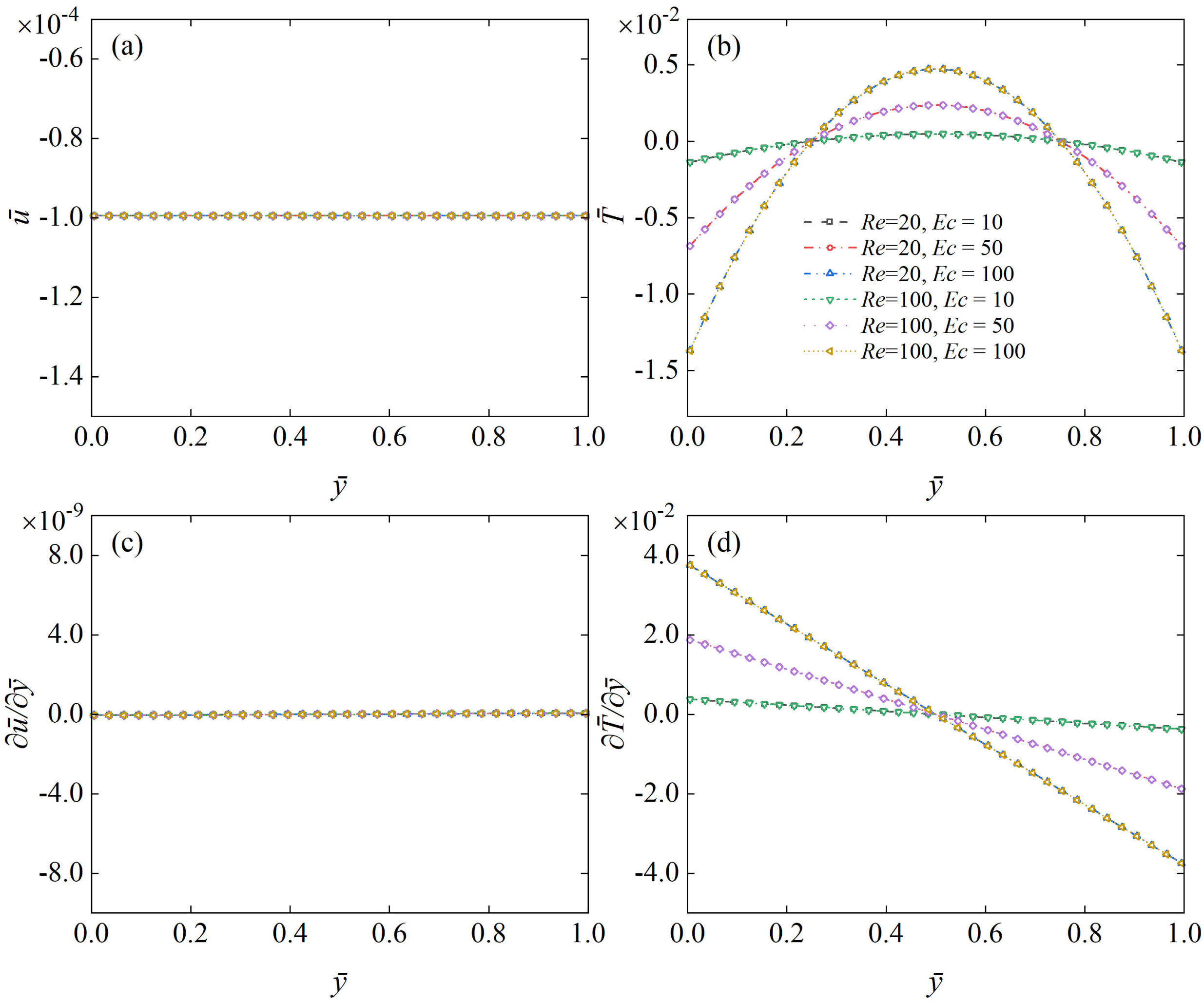

Figure 3: The local error between the analytical and numerical solutions of (a) the horizontal velocity $\bar{u}$, (b) the temperature $\bar{T}$, (c) the velocity gradient $\partial\bar{u}/\partial\bar{y}$, and (d) the temperature gradient $\partial\bar{T}/\partial\bar{y}$ at Pr = 0.71 along the vertical midline.

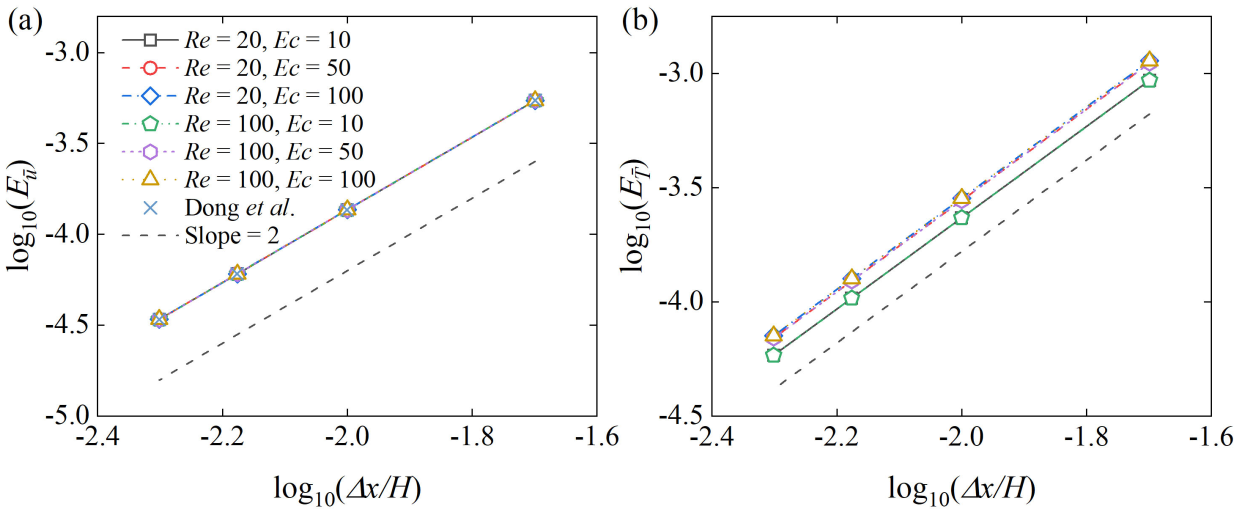

Figure 4: The spatial accuracy of (a) horizontal velocity $\bar{u}$ and (b) temperature $\bar{T}$ at Pr = 0.71. The mesh resolutions 50×50, 100×100, 150×150, and 200×200 are examined, respectively. The dashed lines represent the reference with a slope of 2.

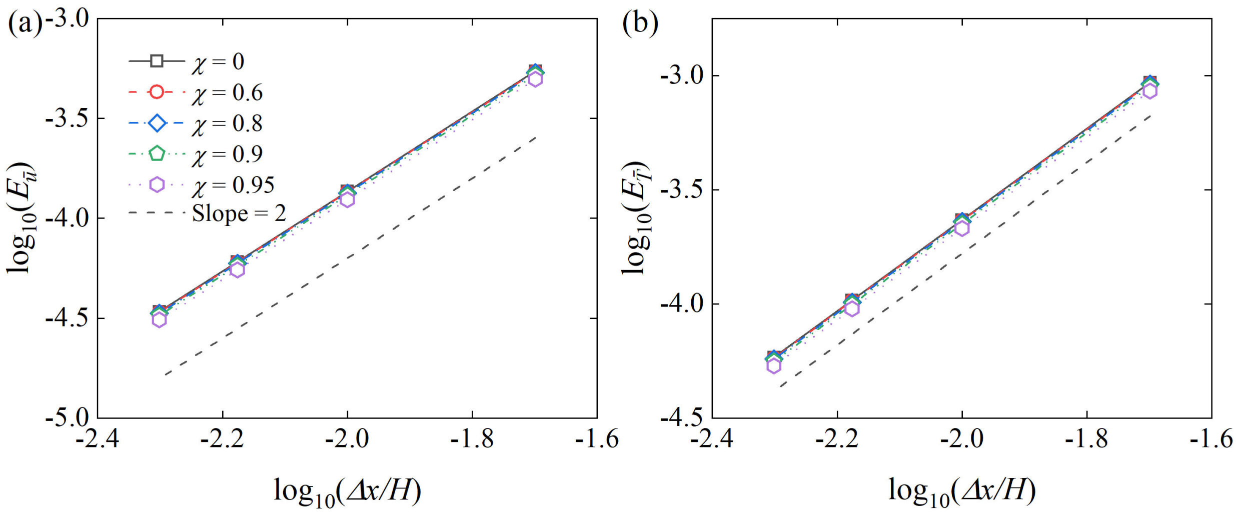

Figure 5: The spatial accuracy of (a) the horizontal velocity $\bar{u}$ and (b) the temperature $\bar{T}$ at Re = 20, Ec = 10, and Pr = 0.71, with various values of χ at 0, 0.6, 0.8, 0.9, and 0.95, respectively. The dashed lines provide the reference with slope = 2.

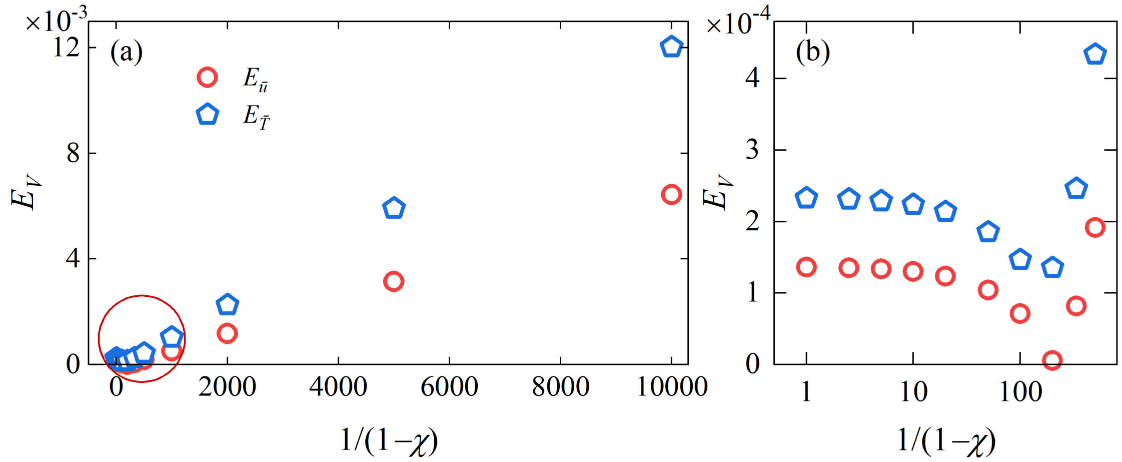

Figure 6: (a) Relative errors for $\bar{u}$ and temperature $\bar{T}$ with the variation of χ, and (b) a zoomed view of the data within the red circle.

For the relative error in scenarios involving χ_s and χ_κ, variations in these two free parameters are considered under the conditions of $Re = 20$, $Ec = 10$, and $Pr = 0.71$. To simplify and facilitate the analysis, the simultaneous change $\chi_s = \chi_\kappa = \chi$ is employed, with the values $\chi = 0$, 0.6, 0.8, 0.9, 0.95. We utilize a viscosity of $\nu = 0.01$ and a corresponding diffusivity of $\kappa = \nu/Pr = 0.014$ in LBM units, and the results are illustrated in Fig. 5. Although the variation of χ slightly influences the relative errors in velocity and temperature, it does not affect the second-order accuracy of the present scheme, which is consistent with the results of Wang *et al.* [37] where the relaxation time was varied.

To further investigate the effect of χ on the relative error, a series of χ values, ranging from 0 to 0.9999 (represented by $1/(1-\chi)$, the magnification factor of the default relaxation times τ_f and τ_g, ranging from 1 to 10000), is employed in the scenarios with $Re = 20$, $Ec = 10$, and $Pr = 0.71$, along with the default $\tau_{fL} = 0.53$. The results for $\bar{u}$ and $\bar{T}$ are presented in Fig. 6. In general, the relative error significantly increases with increasing $1/(1-\chi)$ for the monitored variables $\bar{u}$ and $\bar{T}$. It is observed that the error for temperature $\bar{T}$ exceeds 0.01 at $1/(1-\chi) = 10000$, although the simulation remains stable throughout the reported scenarios. Comparatively, within the range of $1/(1-\chi) = 1 \sim 500$, the temperature error initially decreases marginally as $1/(1-\chi)$ increases and rises rapidly after reaching a minimum error of ($E_{\bar{T}} = 1.36 \times 10^{-4}$) at $1/(1-\chi) = 200$. A similar trend is detected for the horizontal velocity error, which exhibits the minimum value of $E_{\bar{u}} = 5.48 \times 10^{-6}$ at the same $1/(1-\chi)$. The error is considered acceptable for this problem when $1/(1-\chi) < 333.33$ (i.e., $\chi < 0.997$), with even better performance noted when $1/(1-\chi)$ is within 20.0 owing to minimal variations.

From Eqs. 27 and 29, the relationships between the LBM relaxation times for the flow and scalar fields and their corresponding tunable parameters, χ_s and χ_κ, are obtained as

$$\tau_f = \left(\tau_{fL} - 0.5\right) dt = \frac{\nu}{c_s^2}\frac{1}{1-\chi_s}; \; \tau_g = \left(\tau_{gL} - 0.5\right) dt = \frac{\kappa}{c_s^2}\frac{1}{1-\chi_\kappa}, \tag{47}$$

which directly reveals that increasing tunable parameter χ (representing both χ_s and χ_κ), i.e., in-

Table 1: Relative error for thermal Poiseuille flow at $Re = 20$ and $Ec = 10$, compared with previously proposed LKS-like schemes.

χ_s		$E_{\bar{u}}$	$E_{\bar{T}}$	$E_{\partial\bar{u}/\partial\bar{y}}$	$E_{\partial\bar{T}/\partial\bar{y}}$
0	Wang [38]	8.173×10^{-4}	2.920×10^{-3}	9.349×10^{-3}	3.972×10^{-2}
	Yang [39]	1.501×10^{-4}	-	-	-
	Present	1.116×10^{-4}	1.204×10^{-3}	2.146×10^{-13}	7.413×10^{-4}
0.2	Wang [38]	2.653×10^{-4}	2.035×10^{-3}	9.349×10^{-3}	3.972×10^{-2}
	Present	2.008×10^{-4}	1.431×10^{-3}	2.146×10^{-13}	7.417×10^{-4}
0.4	Wang [38]	5.307×10^{-4}	2.523×10^{-3}	9.349×10^{-3}	3.972×10^{-2}
	Present	2.899×10^{-4}	1.660×10^{-3}	2.025×10^{-13}	7.424×10^{-4}
0.8	Wang [38]	1.061×10^{-3}	3.575×10^{-3}	9.349×10^{-3}	3.972×10^{-2}
	Present	4.683×10^{-4}	2.122×10^{-3}	1.885×10^{-13}	7.446×10^{-4}

creasing $1/(1-\chi)$, effectively raises the physical relaxation time τ (representing both τ_f and τ_g), push the LBM relaxation time τ_L (representing both τ_{fL} and τ_{gL}) away from 0.5 at fixed low viscosity/diffusivity. Accordingly, increasing the relaxation time amplifies truncation errors, since it is one of the main contributors of errors in LBM [51–53], as also reflected in Eqs. (9) and (20), consistent with the trends shown in Fig. 6. Consequently, the practical choice of χ should follow the guidelines for the relaxation time, using τ_L as an illustrative example. Both the report by Holdych et al. [53] and the results shown in Fig. 6 indicate that $\tau_L \gg 1$ (i.e., $1/(1-\chi) \gg 1$ in this study) should be avoided. Specifically, the error in Fig. 6 remains comparable to that of the default scheme when $1/(1-\chi) \leq 200$, corresponding to $\tau_{fL} \leq 6.5$ and $\tau_{gL} \leq 8.5$ ($\tau_{fL}^{\text{defalut}} = 0.53$ and $\tau_{gL}^{\text{defalut}} = 0.54$).

To further demonstrate the accuracy of the present scheme, a series of comparisons with previously proposed LKS-like schemes [38, 39] are conducted to highlight its superiority, considering both the default settings and variations in the parameters χ_s and χ_κ. These comparisons are performed under the same conditions as those reported in the literature, at $Re = 20$ and $Ec = 10$, with the LBM relaxation time τ_{fL} fixed at 1, and the channel domain discretized using a 64×64 mesh. For consistency with the referred references, χ_κ is set differently from χ_s in this part, following the relation $\chi_\kappa = Pr\chi_s$. The relative errors, compared with those reported in the literature, are listed in Table 1. In addition to the previous agreement with the theoretically predicted hidden error from the bounce-back boundary [47], the results in Table 1 further demonstrates that the present scheme achieves better accuracy.

3.2. Thermal Couette flow

Thermal Couette flow serves as another commonly used validation problem for thermal flow studies. Its schematic is illustrated in Fig. 7. In this configuration, the upper wall, maintained at a temperature of $T = T_h$, moves rightwards at a velocity of u_0, while the lower wall, at a temperature of $T = T_c$, remains stationary. The heat source term Q, resulting from heat dissipation, is also

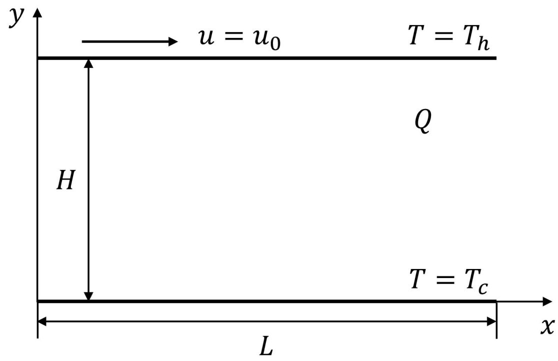

Figure 7: Schematic of the thermal Couette flow

defined as Eq. (42). This problem is characterized by dimensionless parameters Re, Pr, and Ec, with their formulations identical to those expressed in Eq. (43).

The analytical solutions for the dimensionless velocity $(\bar{u}, \bar{v}) = (u, v)/u_0$ and temperature $\bar{T} = (T - T_c)/\Delta T$ in this problem, are expressed as

$$\bar{u} = \bar{y}, \bar{v} = 0, \tag{48a}$$

$$\bar{T} = \bar{y} + \frac{PrEc}{2}\bar{y}\left(1 - \bar{y}\right), \tag{48b}$$

and their gradients are given by

$$\frac{\partial \bar{u}}{\partial \bar{y}} = 1, \tag{49a}$$

$$\frac{\partial \bar{T}}{\partial \bar{y}} = \Delta T \left[1 + \frac{1}{2} PrEc\left(1 - 2\bar{y}\right)\right]. \tag{49b}$$

In this problem, the investigation considers the scenarios at a fixed Reynolds number of $Re = 20$ and Prandtl number of $Pr = 0.71$, while varying the Eckert numbers at $Ec = 10$, 50, and 100. The default value of τ_{fL} is 0.53, and the variation of the two free parameters is also synchronized by $\chi_s = \chi_\kappa = \chi$. The moving upper wall and stationary lower wall are implemented using the halfway bounce-back scheme, while the isothermal boundaries are realized through the anti-bounce-back scheme. The left and right boundaries exhibit periodicity in both the velocity and temperature fields. The computational domain has an aspect ratio of $H/L = 1.0$ and a resolution of 100×100.

Figure 8 provides a comparison of the dimensionless velocity $\bar{u}$ and temperature $\bar{T}$, together with their respective gradients, against the analytical solutions along the midline $x/L = 0.5$. The results demonstrate close agreement for each scenario. Fig. 9 depicts the local error of the four physical variables ($\bar{u}$, $\bar{T}$, $\partial\bar{u}/\partial\bar{y}$, and $\partial\bar{T}/\partial\bar{y}$), at different values of Ec (10, 50, 100) and χ (0 and 0.8). The variations in Ec and χ have no impact on the local error for the velocity, as both $\bar{u}$ and $\partial\bar{u}/\partial\bar{y}$ show no error along $x/L = 0.5$, as illustrated in Figs. 9 (a) and (c), respectively. In contrast, the local error for $\bar{T}$ remains constant along the y-coordinate, which can be viewed as a temperature slip due to the hidden error in boundary condition implementation. Such hidden error was analyzed analytically in Dong *et al.* [47] for the velocity field, which shows that this

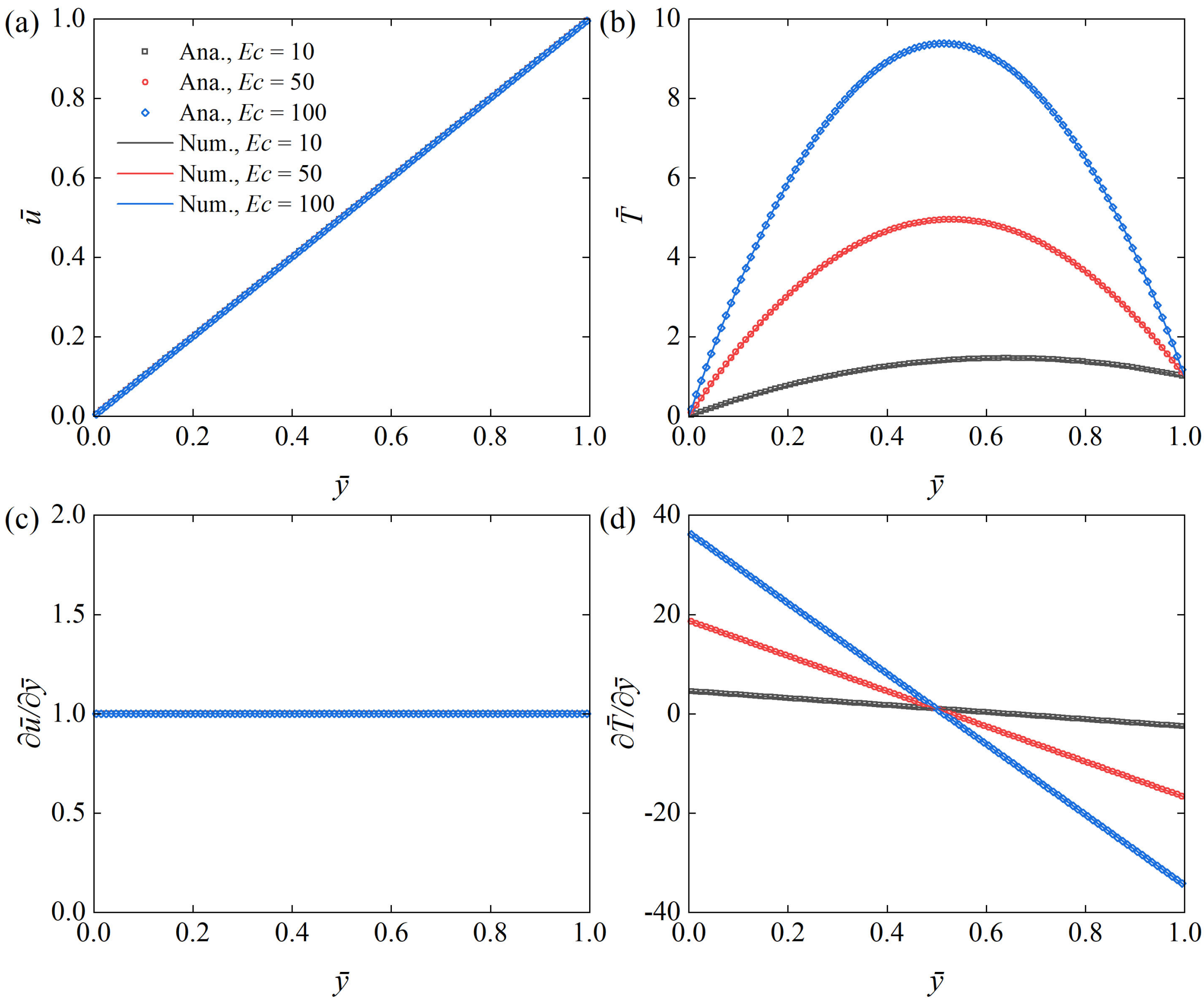

Figure 8: The profile comparison between the analytical and numerical solution of (a) the horizontal velocity $\bar{u}$, (b) the temperature $\bar{T}$, (c) the velocity gradient $\partial\bar{u}/\partial\bar{y}$, and (d) the temperature gradient $\partial\bar{T}/\partial\bar{y}$ at Pr = 0.71 and Re = 20 along the vertical midline.

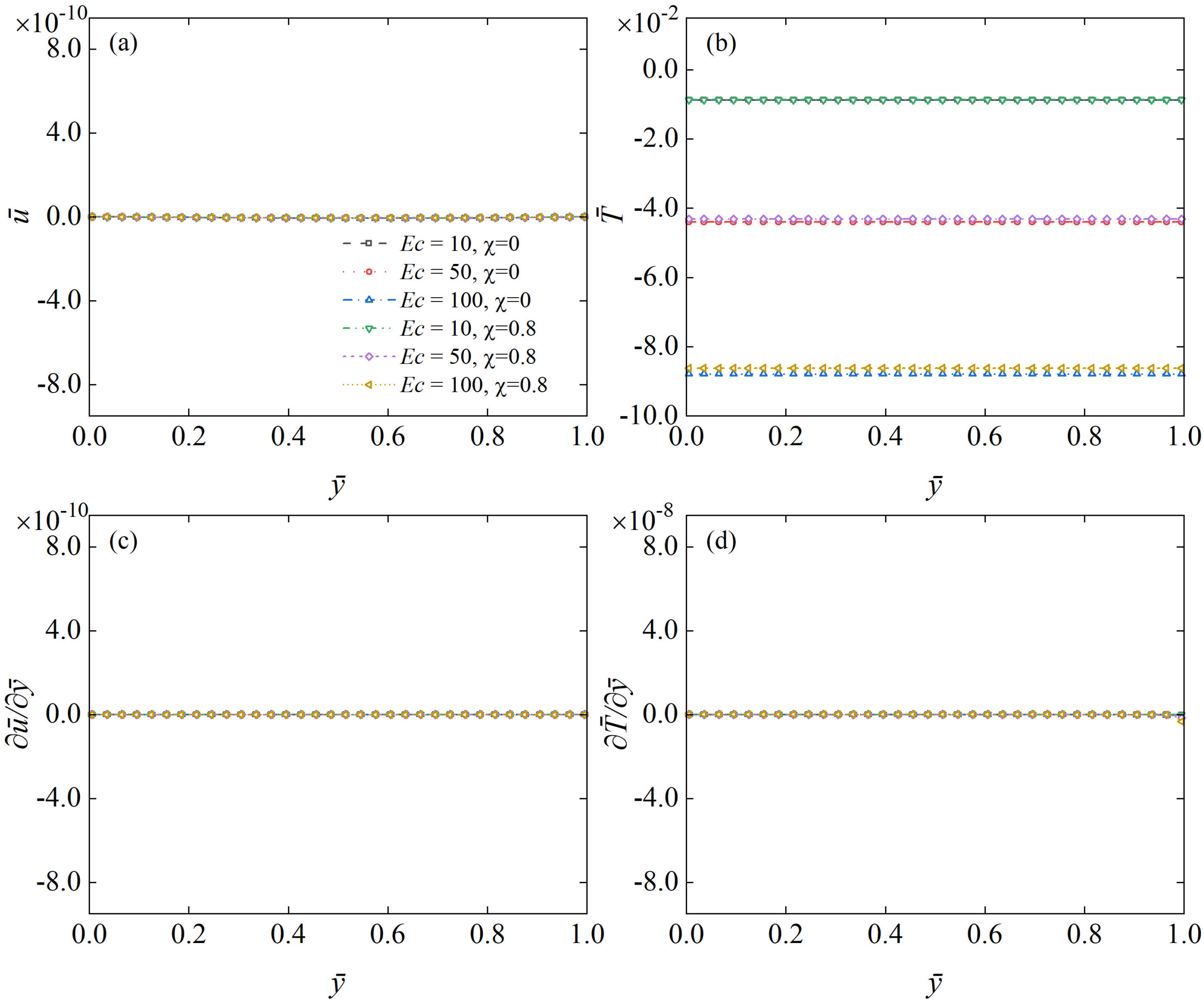

Figure 9: The local error between the analytical and numerical solutions of (a) the horizontal velocity $\bar{u}$, (b) the temperature $\bar{T}$, (c) the velocity gradient $\partial\bar{u}/\partial\bar{y}$, and (d) the temperature gradient $\partial\bar{T}/\partial\bar{y}$ at Pr = 0.71 and Re = 20 along the vertical midline.

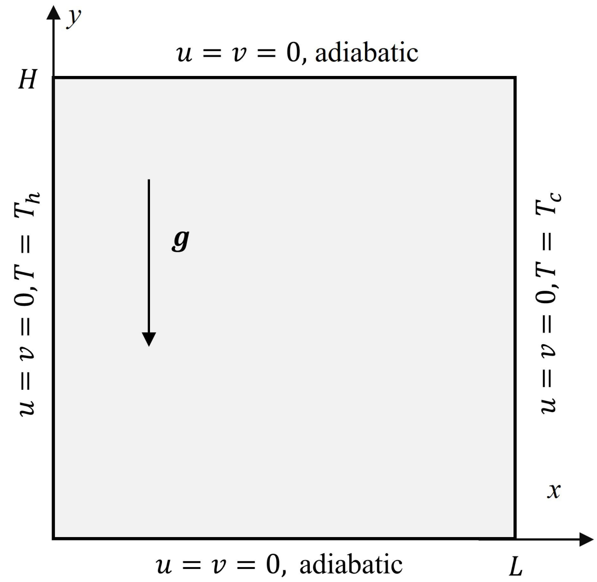

Figure 10: Schematic of the nature convection problem

hidden error causes a velocity slip to be proportional to the local second normal derivative of the velocity. Therefore, it is reasonable to expect the temperature slip due to the hidden error is also proportional to the local second derivative of temperature. Then Eq. (48b) implies the temperature slip is proportional to Ec. Fig. 9 (b) shows that this is indeed the case.

3.3. Natural convection problem

The natural convection problem is a buoyancy-driven thermal flow within an enclosure where all boundaries are fixed. As depicted in Fig. 10, the cavity has dimensions $L \times H$ with an aspect ratio of $H/L = 1$. The two horizontal walls are adiabatic, while the left wall is heated ($T = T_h$) and the right wall is cooled ($T = T_c$). Here, we define $\Delta T = T_h - T_c$ to represent the temperature difference, and $T_0 = (T_h + T_c)/2$ to denote the reference temperature. The Boussinesq approximation [54] is utilized, with the external buoyancy force per unit mass given by

$$F = g\beta(T - T_0)\boldsymbol{j}, \tag{50}$$

where g represents gravitational acceleration, and β denotes the thermal expansion coefficient. $\boldsymbol{j}$ signifies the unit vector along the y-axis. This natural convection problem is characterized by the Prandtl number Pr and Rayleigh number Ra, and they are defined as

$$Pr = \frac{\nu}{\kappa}, \quad Ra = \frac{g\beta\Delta T L^3}{\nu\kappa}. \tag{51}$$

The reference velocity and time are defined as $u_0 = \sqrt{g\beta\Delta T H}$ and $t_0 = H/u_0$, respectively. Due to the symmetric properties of the configuration, the average Nusselt number for the vertical wall is represented by the value at the cold wall and is calculated as

$$\overline{Nu} = \int_0^H Nu(L, y)dy/H, \tag{52}$$

where $Nu(L,y) = -\frac{L}{\Delta T}\left(\frac{\partial T}{\partial x}\right)_{\text{cold wall}}$, and $\partial T/\partial x$ can be locally obtained using Eq. (35). Meanwhile, a maximum Nusselt number $Nu_{\max}$ at the vertical midline is defined as the maximum value of $Nu_{\max}(x = 0.5L) = -\frac{L}{\Delta T}\max\left[\left(\frac{\partial T(x=0.5L,y)}{\partial x}\right)\right]$.

Table 2 compares the present solution with the literature benchmark data, including the scenarios with $\chi_s = \chi_\kappa = \chi = 0$ and 0.8. The Prandtl number is held constant at $Pr = 0.71$, while various Rayleigh numbers are considered, specifically, $Ra = 10^5, 10^6, 10^7, 10^8, 10^9$, and 10^{10}. Correspondingly, the mesh resolutions with uniform grid vary with the Rayleigh number, as 256^2, 256^2, 512^2, 512^2, 1024^2, and 2048^2, respectively. These spatial meshes are capable of adequately resolving the thermal boundary layer [55–58]. The relaxation times τ_{fL} are set to 0.6019, 0.5322, 0.5204, 0.5065, 0.5082, and 0.5104, respectively for $\chi = 0$, and 1.0096, 0.6612, 0.6021, 0.5323, 0.5409, and 0.5518, respectively for $\chi = 0.8$. Since this vertical convection thermal flow becomes unsteady at the critical Rayleigh number $Ra = 1.82 \times 10^8$ [59], the results for unsteady flows ($Ra = 10^9$ and 10^{10}) are averaged over time once the system becomes stationary. The time averaging is conducted over 100 instants with a dimensionless time interval of $u_0\Delta t/H = 5$.

Table 2 presents several characteristic flow properties: the maximum horizontal velocity u_{max} at the vertical midline $x = 0.5L$ and its corresponding location y_{max}, the maximum vertical velocity v_{max} at the horizontal midline $y = 0.5H$ and its location x_{max}, $Nu_{max}(x = 0.5L)$ and its corresponding location $y_{\max}$, and the average Nusselt number $\overline{Nu}$ at the cold wall. It is important to note that all velocity and location variables in the table have been normalized by the reference velocity u_0, and the reference length H. The average Nusselt number $\overline{Nu}$ differs typically by only 1.1%, except for the $Ra = 10^{10}$ case where the relative difference is 1.5%. These results demonstrate a good agreement with the literature data for each investigated Rayleigh number Ra and free parameter χ.

Figure 11 presents the isotherms for various Rayleigh numbers ranging from 10^5 to 10^{10}, using $\chi = 0$ as the example, as the results are identical for $\chi = 0.8$. It is evident that the thermal boundary layer near the vertical walls gradually becomes thinner as the Rayleigh number increases, which aligns with the observations made by Zekeriya [62] and Wang [63]. Figure 12 shows the relationship between the thermal boundary layer thickness and the increasing Rayleigh number. The thickness is determined at the wall-normal location where the conductive heat flux $-\kappa d\langle T\rangle/dx$ equals the convective heat flux $\langle uT\rangle$, along the horizontal direction [64]. Results reveal that the thickness agrees roughly with the theoretical scaling law of $\delta_T/L \sim Ra^{-1/4}$ for a steady unbounded vertical thermal boundary layer [64, 65]. The actual scaling exponent appears to be -0.277. This slight departure represents the effect of the wall and unsteady flow at high Ra numbers.

To examine the enhancement of numerical stability, we consider a scenario with $Ra = 10^{10}$ and $Pr = 0.71$ as an example, reducing L to create conditions with low viscosity ν and diffusivity κ as described in Eq. (51). Figure 13 presents the temperature and horizontal velocity fields for the scenario with $\nu = 8.43 \times 10^{-5}$ and $\kappa = 1.19 \times 10^{-4}$ (the corresponding relaxation times in flow and temperature fields for the baseline model ($\chi = 0$) are $\tau_{fL} = 0.5 + 2.53 \times 10^{-4}$ and $\tau_{gL} = 0.5 + 3.57 \times 10^{-4}$, respectively), at a dimensionless time $t = 2.9$, which is the instant just before the occurrence of unphysical results due to numerical instability. Visible numerical oscillations are observed in the top-left and bottom-right corners of the temperature field, together with the numerical noises near the upper left wall and the lower right wall in the horizontal velocity field. However, adjusting the free parameter χ to 0.9 without changing the other parameters, effectively mitigates the numerical instability problem, resulting in relaxation times for flow and

Table 2: Comparison of the present results (including $\chi = 0$ and $\chi = 0.8$) with the reference benchmark solutions.

Ra		u_{max}	y_{max}	v_{max}	x_{max}	Nu_{max}	y_{Nu}	$\overline{Nu}$
10^5	Yu [29]	0.1288	0.8550	0.2552	0.0660	-	-	4.508
	Liu [60]	0.1303	0.8550	0.2569	0.0650	7.743	0.9200	4.520
	$\chi = 0$	0.1305	0.8477	0.2569	0.0625	7.718	0.9141	4.522
	$\chi = 0.8$	0.1302	0.8471	0.2569	0.0627	7.720	0.9137	4.522
10^6	Yu [29]	0.0756	0.8520	0.2582	0.0400	-	-	8.737
	Liu [60]	0.0773	0.8500	0.2604	0.0400	17.59	0.9600	8.773
	$\chi = 0$	0.0769	0.8438	0.2610	0.0352	17.48	0.9570	8.824
	$\chi = 0.8$	0.0768	0.8431	0.2610	0.0353	17.49	0.9529	8.825
10^7	Xu [56]	0.0558	0.8791	0.2625	0.0213	-	-	16.52
	Contrino [61]	0.0558	0.8793	0.2625	0.0213	-	-	16.52
	$\chi = 0$	0.0558	0.8789	0.2621	0.0195	39.20	0.9805	16.49
	$\chi = 0.8$	0.0557	0.8767	0.2618	0.0196	39.24	0.9785	16.52
10^8	Xu [56]	0.0381	0.9279	0.2634	0.0119	-	-	30.23
	Contrino [61]	0.0381	0.9279	0.2632	0.0120	-	-	30.23
	$\chi = 0$	0.0380	0.9256	0.2620	0.0117	85.32	0.9883	30.18
	$\chi = 0.8$	0.0382	0.9256	0.2620	0.0117	85.79	0.9883	30.17
10^9	Sharma [55]	0.0184	0.911	0.2640	0.0064	-	-	54.82
	Quéré [59]	-	-	-	-	-	-	54.60
	$\chi = 0$	0.0187	0.9032	0.2632	0.0059	184.5	0.9941	54.60
	$\chi = 0.8$	0.0178	0.9014	0.2609	0.0068	174.9	0.9961	54.00
10^{10}	Sharma [55]	0.0148	0.97	0.2644	0.0034	-	-	100.2
	Quéré [59]	-	-	-	-	-	-	100.0
	$\chi = 0$	0.0115	0.9726	0.2652	0.0034	398.89	0.9976	99.21
	$\chi = 0.8$	0.0118	0.9868	0.2635	0.0039	376.03	0.9966	98.46

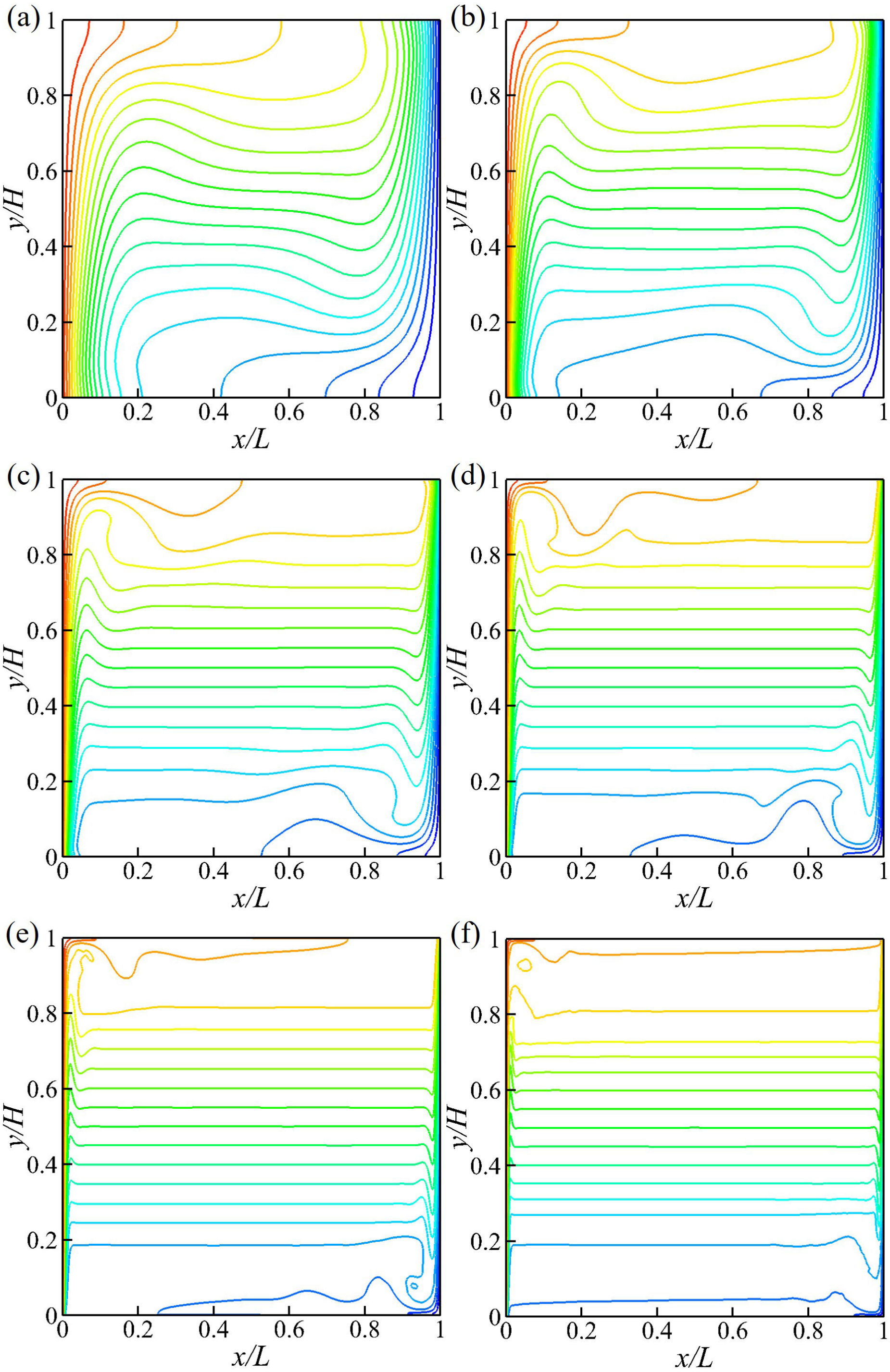

Figure 11: Isotherms for various Rayleigh number (a) $Ra = 10^5$, (b) $Ra = 10^6$, (c) $Ra = 10^7$, (d) $Ra = 10^8$, (e) $Ra = 10^9$, and (f) $Ra = 10^{10}$ at fixed Prandtl number $Pr = 0.71$.

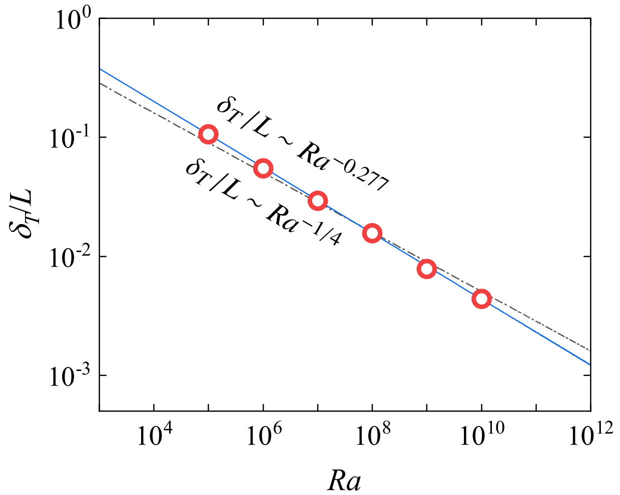

Figure 12: The relationship between the thermal boundary layer thickness (δ_T/L) and Rayleigh number (Ra). The blue solid line denotes the scaling law for thermal boundary layer thickness, with $\delta_T/L \sim Ra^{-0.277}$, alongside a blue solid line for the theoretical solution for plate thermal boundary layer thickness, given by $\delta_T/L \sim Ra^{-1/4}$.

temperature fields of $\tau_{fL} = 0.5 + 2.53 \times 10^{-3}$ and $\tau_{gL} = 0.5 + 3.57 \times 10^{-3}$, respectively.

For a more complete demonstration of the enhanced numerical stability of our model, a series of scenarios with various viscosity values ν and the corresponding diffusivity κ, have been examined, at $Ra = 10^8$, 10^9, and 10^{10}. The variation of ν is achieved by adjusting L, and the numerical stability regime diagram are presented in Fig. 14. When $\chi = 0.0$, the simulation becomes diverging as ν decreases to approximately 2.0×10^{-4} for each Rayleigh number, as indicated by the detection of "*NaN*" values in the velocity and temperature fields. However, the numerical stability is restored when χ is adjusted to 0.9, with code divergence occurring only when ν falls below 1.0×10^{-4} $[L^2T^{-1}]$. All tested cases are stable with a further increase of χ to 0.95. As the revised τ_f and τ_g are $1/(1-\chi)$ times their respective default values, $\chi = 0.9$ indicates a tenfold increase, while $\chi = 0.95$ specifies a twentyfold increase relative to the default relaxation time. Increasing χ drives the relaxation time τ_{fL} and τ_{gL} away from 0.5 at low viscosity and diffusivity, thereby significantly enhancing the numerical stability.

3.4. Three-dimensional Rayleigh-Bénard convection

To further verify the improved numerical stability of the proposed scheme and examine the influence of tunable parameters on the numerical accuracy when simulating 3D turbulent flows, 3D Rayleigh-Bénard convection is considered in this subsection, which is a paradigmatic buoyancy-driven flow confined within a stationary cavity. As shown in Fig. 15, the upper wall is maintained at $T_c = 0$ (cold) and the lower wall at $T_h = 1$ (hot); the vertical walls are adiabatic. The cavity dimensions are $L \times L/2 \times H$, with $H/L = 1$. Similar to the vertical convection problem in Sec. 3.3, Rayleigh-Bénard convection is also characterized by the Prandtl number Pr and Rayleigh number Ra, as expressed in Eq. (51), and the same definitions of the reference velocity and characteristic time are utilized in this section. In addition to the mean Nusselt number at the cold wall ($\overline{Nu}$, similar to Eq. (52), but using the temperature gradient along the upper wall), we examine the

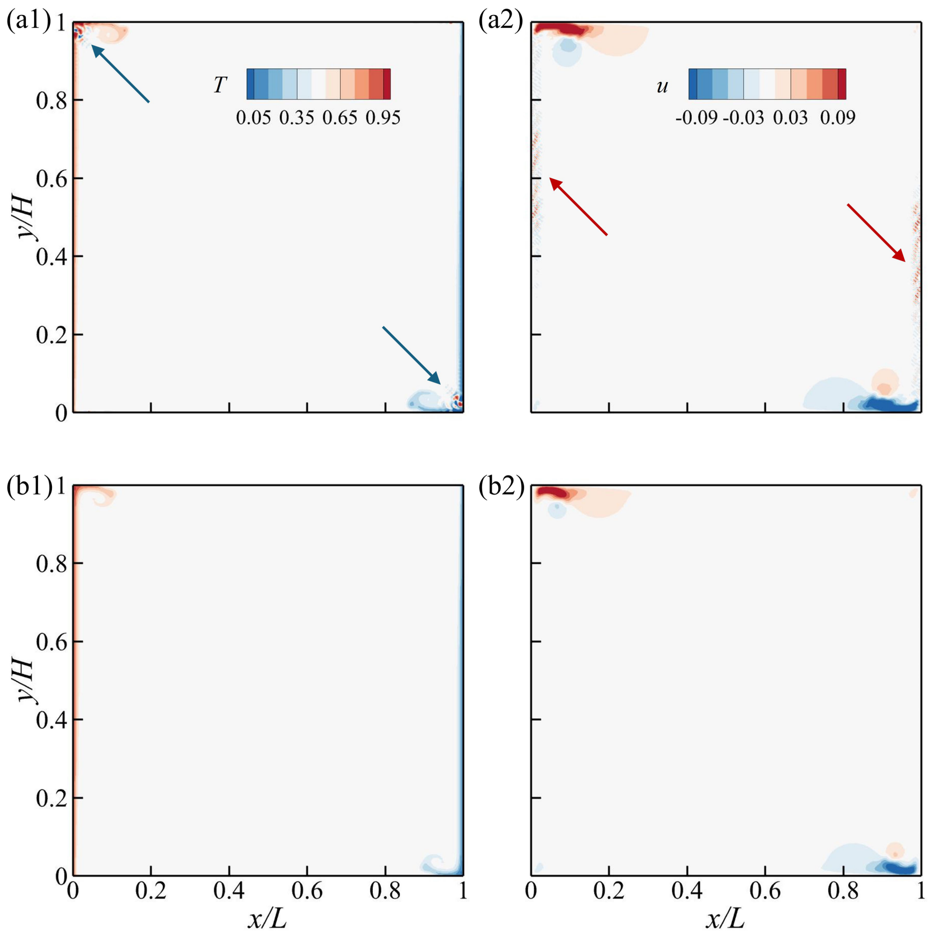

Figure 13: Numerical instability in (a1) the temperature and (a2) horizontal velocity field at dimensionless time $t = 2.9$ for $Ra = 10^{10}$ and $\chi = 0$, as well as the comparison with the corresponding fields for (b1) temperature and (b2) horizontal velocity when $\chi = 0.9$. The blue arrows indicate the oscillation regions for the temperature field, while the red arrows indicate the numerical noises for the velocity field.

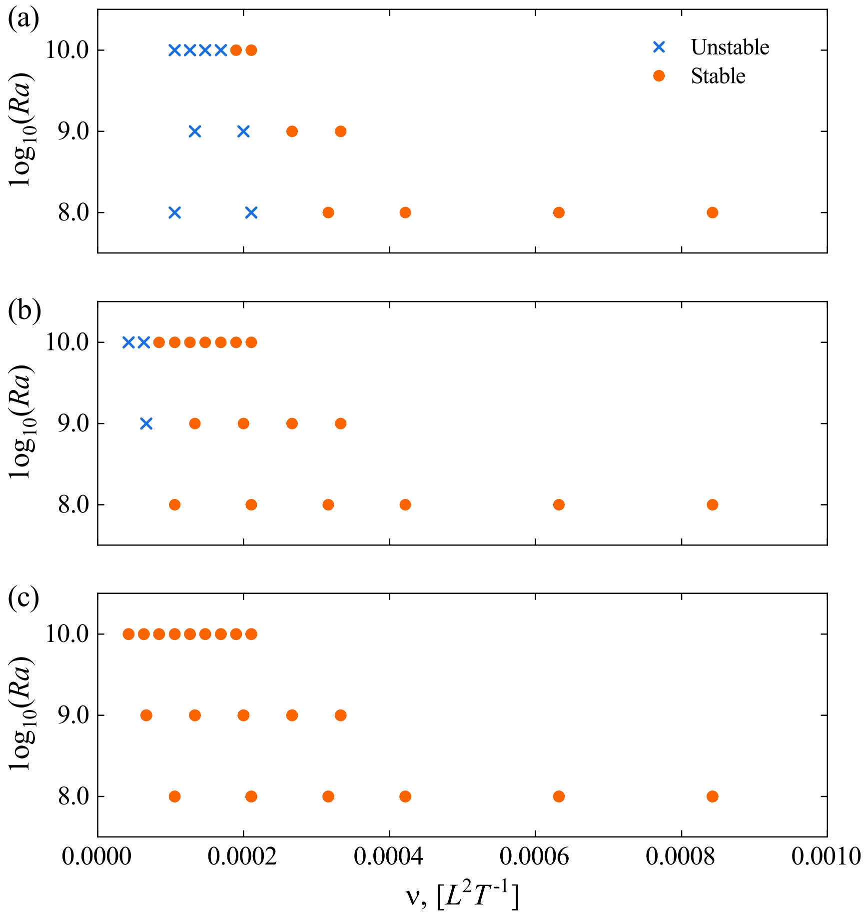

Figure 14: The stability pattern with the variation of free parameter χ. (a) χ = 0.0, (b) χ = 0.9, and (c) χ = 0.95.

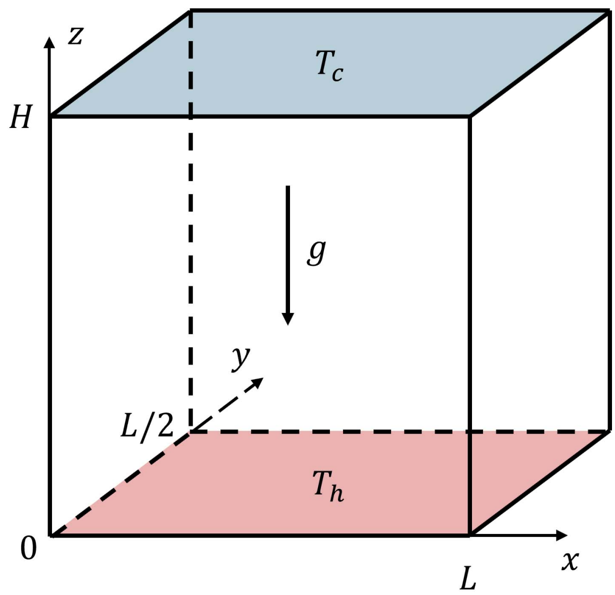

Figure 15: Schematic of the 3D Rayleigh-Bénard convection

volume-averaged Nusselt number in the bulk for comparison, which is defined as

$$\overline{Nu}_B = \frac{\langle vT \rangle}{\kappa \Delta T / H} \tag{53}$$

where $\langle vT \rangle$ denotes the convective heat flux in the vertical direction.

For comparison, the same configuration is also simulated using the finite-difference method (FD). Here we use an open-source code known as AFiD [66], which is third-order accurate in time and second-order accurate in space. A higher resolution with $256 \times 128 \times 256$ uniform grids using FD produces $\overline{Nu} = 8.847$. All the statistics from the FD and present LBM reported are averaged over more than 400 non-dimensional time after the flow has reached its statistically stationary state. Table 3 compares the Nusselt numbers measured at the cold wall ($\overline{Nu}$) and in the bulk flow ($\overline{Nu}_B$)

Table 3: Comparison of Nusselt numbers obtained using three different lattice velocity sets for the scalar field and the varied χ.

Scheme	χ	$\overline{Nu}$	$\overline{Nu}_B$
D3Q15	0	8.788	8.791
	0.5	8.770	8.775
D3Q19	0	8.777	8.779
	0.5	8.768	8.774
D3Q27	0	8.815	8.803
	0.5	8.765	8.769

Table 4: Comparison of mean Nusselt number at the cold wall ($\overline{Nu}$) obtained in 3D Rayleigh-Bénard convection using D3Q19 model for the scalar field with varied χ. Large relative error can be observed when $\chi \geq 0.6$.

χ	0	0.2	0.3	0.5	0.6	0.7	0.8	0.9
$\overline{Nu}$	8.777	8.775	8.774	8.768	8.709	8.606	8.270	7.674
R.E.	0.79%	0.81%	0.83%	0.89%	1.56%	2.72%	6.52%	13.26%

for three lattice velocity sets used in the scalar field (D3Q15, D3Q19, and D3Q27) at $Ra = 10^6$ and $Pr = 0.71$. It should be noted that we only use different velocity sets for the scalar field, while the same D3Q27 model is adopted for the flow field. Two values for χ ($\chi_s = \chi_\kappa = \chi = 0$ and $\chi_s = \chi_\kappa = \chi = 0.5$) are employed. The spatial resolution is $200 \times 100 \times 200$, chosen in strict accordance with the criterion of Shishkina et al. [57], to fully resolve the boundary layer. The results in Table 3 indicate that the Nusselt numbers measured at both the cold wall and in the bulk show negligible deviation for $\chi = 0.5$ compared with $\chi = 0$ in all three velocity sets, demonstrating the accuracy of the present scheme in this 3D configuration. For demonstration purposes, we use the D3Q19 model to examine the influence of χ on numerical accuracy in turbulent thermal flow.

The results using D3Q19 model when varying tunable parameters (χ) are summarized in Table 4, by presenting $\overline{Nu}$ and its relative error (R.E.) to the result of FD ($|(\overline{Nu}_{LBM} - \overline{Nu}_{FD})/\overline{Nu}_{FD}|$). When χ is smaller than 0.5, $\overline{Nu}$ is almost unaffected, and the relative error becomes larger than 1% for $\chi \geq 0.6$. This threshold corresponds to $\tau_L \leq 0.643$ ($\chi \leq 0.5$). Considering the relative errors of D3Q15 and D3Q27 models are comparable to that of the D3Q19 model for $\chi = 0.5$ (less than 1%, see Table 3), we therefore suggest that, for turbulence, τ_L should be kept below approximately 0.65. It should be noted that the parameter ranges suggested here are based on the specific cases considered in this work, and we would like to further explore the influence of the tunable parameters on numerical accuracy and stability in our future work.

Subsequently, numerical stability at low viscosity and diffusivity is examined using coarse-grid simulations. At $Ra = 10^9$ and $Pr = 0.71$, decreasing ν to 3.77×10^{-4} (also achieved by decreasing L to 100, which can not resolve the boundary layer at $Ra = 10^9$ and $Pr = 0.71$) produces a clear divergence in both the temperature and velocity fields for all three lattice-velocity sets. For the D3Q19 case, this divergence is observed at a dimensionless time of $t = 25.24$ (see Fig. 16), with a slice taken at $y/L = 0.46$. The oscillatory regions are concentrated primarily within $0.4 \leq x/L \leq 0.6$ and $0.3 \leq z/H \leq 0.5$ (see Figs. 16 (a1) and (b1)). When $\chi = 0.9$ (τ_L is smaller than 0.65) is employed, the divergence disappears (see Figs. 16 (a2) and (b2)), and the simulation proceeds stably to the prescribed terminal time $t = 2828$. To simply assess the stability of the three lattice velocity sets, we examine the minimum value of χ that yields a convergent simulation under the present conditions. Starting from $\chi = 0$ and increasing in increments of 0.1 until the simulation converges, the minimum convergent χ values for D3Q15, D3Q19, and D3Q27 are found to be 0.9, 0.8, and 0.9, respectively. These results illustrate that D3Q19 exhibits superior numerical stability in the present scheme, which may be attributed to the velocity set distribution despite the same fifth-order GHQ accuracy achieved for these three velocity sets.

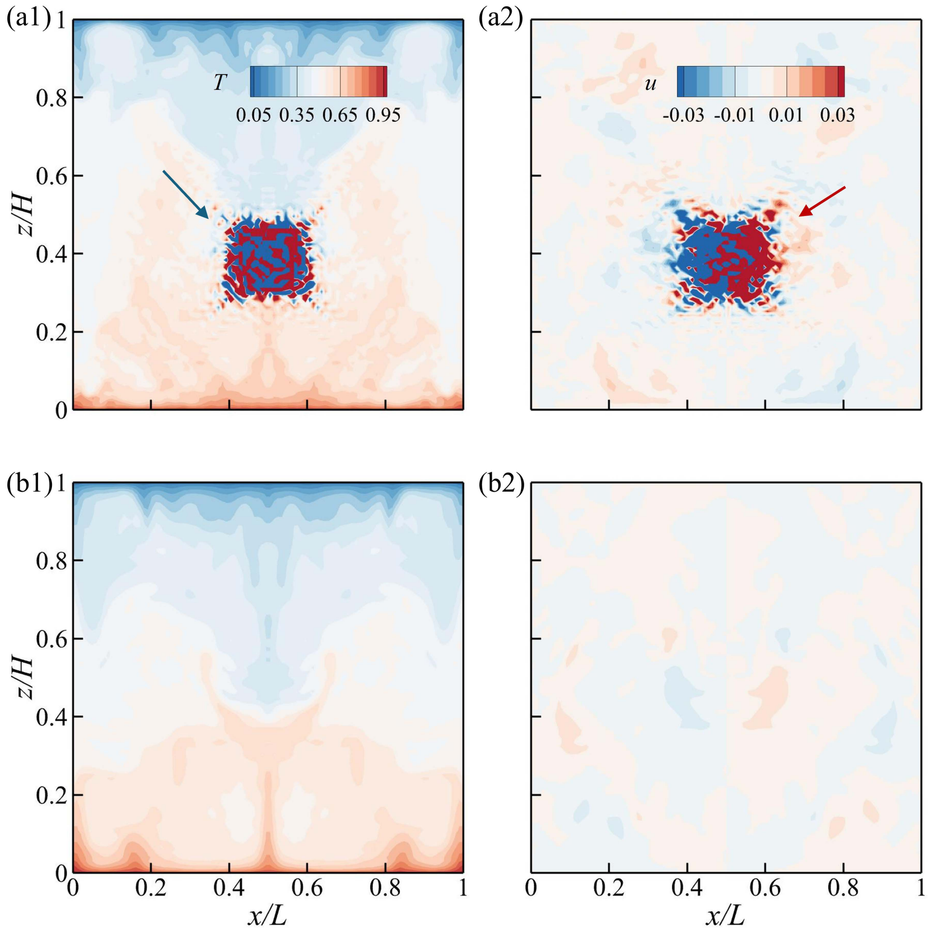

Figure 16: Numerical instability in (a1) the temperature and (a2) velocity along x-axis at dimensionless time $t = 25.24$, in 3D Rayleigh-Bénard convection ($Ra = 10^9$, $Pr = 0.71$, $\chi = 0$, and 200 grids in the vertical direction), and the corresponding converged fields for (a2) temperature and (b2) x-velocity at $\chi = 0.9$, computed with the D3Q19 lattice velocity set, on the plane $y/L = 0.46$. The blue arrow indicates the oscillation regions for the temperature field, while the red arrow indicates the oscillation regions for the velocity field.

4. Summary and Conclusions

In this work, we developed an improved lattice Boltzmann scheme for the convection-diffusion problem through inverse design. The Chapman-Enskog expansion for the continuous Boltzmann-BGK equation is employed to derive the physical constraints on the moments at various orders for the source term, which are then used to design the source term using the Hermite expansion. An adjustable parameter χ is introduced to shift the relaxation time in the LBE away from 0.5 at low viscosity or diffusivity, thereby enhancing numerical stability. This improved scheme can rigorously recover the Navier-Stokes equations and the convection-diffusion equation up to the second order of the Mach number. Furthermore, the strain rate in the flow field and the gradient in the scalar field can be directly calculated from the local non-equilibrium distribution function, thus eliminating any interpolation errors associated with non-local finite differencing.

The developed scheme is first validated using thermal Poiseuille flow and thermal Couette flow, both with viscous dissipation, and shows close agreement with analytical solutions across a range of physical parameters. The local velocity error rigorously follows the theoretical hidden error reported by Dong et al. [47], while the spatial accuracy of both velocity and temperature is second-order and is preserved under variations of the free parameter χ. The comparison with the previous LKS-like schemes demonstrates the superior accuracy of the present scheme. Subsequently, a natural convection problem is used to demonstrate enhanced numerical stability at low viscosity/diffusivity resulting from the increase in χ. The results, covering both laminar and turbulent thermal flow regimes, agree well with the literature in terms of thermal performance and boundary layer thickness. Finally, a 3D Rayleigh-Bénard convection case is examined to validate the accuracy and improved numerical stability of the present scheme, and meanwhile to demonstrate its applicability in the 3D configuration. Increasing χ (i.e., increasing $1/(1-\chi)$) raises the LBM relaxation time τ_L at fixed low viscosity/diffusivity, thereby shifting τ_L away from 0.5 and thus improving numerical stability. However, this enhancement in τ_L amplifies truncation errors of Chapman-Enskog expansion in LBM. Consequently, the choice of χ should follow these guidelines: (1) avoid $1/(1-\chi) \gg 1$, which yields $\tau_L \gg 1$; (2) better to keep $\tau_L \leq 6.5$ for laminar flow and $\tau_L \leq 0.65$ for turbulence.

Despite the modification of the source terms and the introduction of the tunable parameter χ, the proposed model does not substantially increase the complexity of the LBM kernel. Implementation remains similar to the standard LBM and requires only a few additional algebraic calculations of source terms in the collision kernel. Consequently, the scheme retains the computational and implementation advantages of the LBM. Increasing the tunable parameter χ, substantially enhances numerical stability at low viscosity/diffusivity, thereby extending the range of engineering problems that can be addressed in standard LBM, particularly those with large Reynolds and Peclet numbers, and broadening the applicability of the LBM with a simple BGK collision model. However, larger χ values introduce additional truncated errors; therefore, χ must be chosen with care to balance stability and accuracy.

Acknowledgement

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Appendix A. Transformation to lattice Boltzmann equations

This appendix employs the transformation to obtain the LBEs [47], thereby ensuring an efficient derivation of the source terms for LBE. Additionally, the calculation of hydrodynamic variables derived from the transformed distribution function is also presented.

Appendix A.1. The flow filed

The integration of Eq. (3) is performed along the characteristic line on the left-hand side, together with the trapezoidal rule on the right-hand side, as

$$\begin{aligned} f(\boldsymbol{v},\boldsymbol{x}+\boldsymbol{v}dt,t+dt)-f(\boldsymbol{v},\boldsymbol{x},t) &= -\frac{dt}{2\tau_f}\left[f(\boldsymbol{v},\boldsymbol{x}+\boldsymbol{v}dt,t+dt)-f^{eq}(\boldsymbol{v},\boldsymbol{x}+\boldsymbol{v}dt,t+dt)\right] \\ &+\frac{dt}{2}\mathcal{S}(\boldsymbol{v},\boldsymbol{x}+\boldsymbol{v}dt,t+dt)-\frac{dt}{2\tau_f}\left[f(\boldsymbol{v},\boldsymbol{x},t)-f^{eq}(\boldsymbol{x},t)\right]+\frac{dt}{2}\mathcal{S}(\boldsymbol{v},\boldsymbol{x},t)+O(dt^3,\tau_f^2dt). \end{aligned} \tag{A.1}$$

To remove the implicit nature in time, we introduce the following transformation

$$\widetilde{f}(\boldsymbol{v},\boldsymbol{x},t)=f(\boldsymbol{v},\boldsymbol{x},t)+\frac{dt}{2\tau_f}\left[f(\boldsymbol{v},\boldsymbol{x},t)-f^{eq}(\boldsymbol{v},\boldsymbol{x},t)\right]-\frac{dt}{2}\mathcal{S}(\boldsymbol{v},\boldsymbol{x},t). \tag{A.2}$$

Substituting Eq. (A.2) into Eq. (A.1) and ignoring $O(dt^3,\tau_f^2dt)$, one can obtain the explicit LBE as

$$\widetilde{f}(\boldsymbol{v},\boldsymbol{x}+\boldsymbol{v}dt,t+dt)-\widetilde{f}(\boldsymbol{v},\boldsymbol{x},t)=-\frac{1}{\tau_{fL}}\left[\widetilde{f}(\boldsymbol{v},\boldsymbol{x},t)-f^{eq}(\boldsymbol{v},\boldsymbol{x},t)\right]+\Phi(\boldsymbol{v},\boldsymbol{x},t)dt, \tag{A.3}$$

where $\tau_{fL}=\tau_f/dt+0.5$ is the relaxation time in LBM. The source term $\Phi(\boldsymbol{v},\boldsymbol{x},t)$ in LBE is expressed as

$$\Phi(\boldsymbol{v},\boldsymbol{x},t)=\left(1-\frac{1}{2\tau_{fL}}\right)\mathcal{S}(\boldsymbol{v},\boldsymbol{x},t). \tag{A.4}$$

The hydrodynamic variables, $\delta\rho$, u_α, and $\sigma_{\alpha\beta}$, are related to the moments of f (i.e., $f(\boldsymbol{v},\boldsymbol{x},t)$) in the continuous Boltzmann equation. Since $\widetilde{f}$ (i.e., $\widetilde{f}(\boldsymbol{v},\boldsymbol{x},t)$) is actually solved in LBE, we need to establish the relations between the hydrodynamic variables and the moments of $\widetilde{f}$. Using the inverse transformation of Eq. (A.2), as

$$f(\boldsymbol{v},\boldsymbol{x},t)=\left(1-\frac{1}{2\tau_{fL}}\right)\widetilde{f}(\boldsymbol{v},\boldsymbol{x},t)+\frac{1}{2\tau_{fL}}f^{eq}(\boldsymbol{v},\boldsymbol{x},t)+\left(1-\frac{1}{2\tau_{fL}}\right)\frac{dt}{2}\mathcal{S}(\boldsymbol{v},\boldsymbol{x},t), \tag{A.5}$$

these hydrodynamic variables can be obtained by the known moments of f_α^{eq} and S_α. First, consider the zeroth and first-order moments and substitute Eq. (A.5), we have

$$\delta\rho \equiv \int f d\boldsymbol{v} = \left(1 - \frac{1}{2\tau_{fL}}\right)\int \widetilde{f} d\boldsymbol{v} + \frac{1}{2\tau_{fL}}\delta\rho, \tag{A.6a}$$

$$\rho_0 u_\alpha \equiv \int v_\alpha f d\boldsymbol{v} = \left(1 - \frac{1}{2\tau_{fL}}\right)\int v_\alpha \widetilde{f}(\boldsymbol{x}, t) d\boldsymbol{v} + \frac{1}{2\tau_{fL}}\rho_0 u_\alpha + \left(1 - \frac{1}{2\tau_{fL}}\right)\frac{dt}{2}F_\alpha. \tag{A.6b}$$

Simplifying the above two equations yields

$$\delta\rho = \int \widetilde{f} d\boldsymbol{v}, \tag{A.7a}$$

$$\rho_0 u_\alpha = \int v_\alpha \widetilde{f}(\boldsymbol{x}, t) d\boldsymbol{v} + \frac{dt}{2}F_\alpha. \tag{A.7b}$$

Next, the same process is applied to the viscous stress, namely,

$$\sigma_{\alpha\beta} = -\int (v_\alpha - u_\alpha)(v_\beta - u_\beta)\left(f - f^{eq}\right) d\boldsymbol{v} = -\int v_\alpha v_\beta \left(f - f^{eq}\right) d\boldsymbol{v}. \tag{A.8}$$

By substituting Eq. (A.5) into Eq. (A.8), we obtain

$$\sigma_{\alpha\beta} = -\int v_\alpha v_\beta \left(\widetilde{f} - f^{eq}\right) d\boldsymbol{v} - \frac{dt}{2}\left(u_\alpha F_\beta + u_\beta F_\alpha + 2\rho_0 c_s^2 S_{\alpha\beta}\right). \tag{A.9}$$

Next, under the Stokes hypothesis we substitute for $S_{\alpha\beta}$ using the following relations

$$\sigma_{\alpha\beta} = 2\mu\left(S_{\alpha\beta} - \frac{1}{D}S_{\gamma\gamma}\delta_{\alpha\beta}\right) + \mu^B S_{\gamma\gamma}\delta_{\alpha\beta}, \tag{A.10a}$$

$$\sigma_{\gamma\gamma} = D\mu^B S_{\gamma\gamma}. \tag{A.10b}$$

Thus, we have

$$S_{\alpha\beta} = \frac{\sigma_{\alpha\beta}}{2\mu} + \left(\frac{1}{D} - \frac{\mu^B}{2\mu}\right)\frac{\sigma_{\gamma\gamma}}{D\mu^B}\delta_{\alpha\beta}. \tag{A.11}$$

This transforms Eq. (A.9) into

$$\sigma_{\alpha\beta} + dt\rho_0 c_s^2 \frac{\sigma_{\alpha\beta}}{2\mu} + \left(\frac{1}{D} - \frac{\mu^B}{2\mu}\right)\frac{\sigma_{\gamma\gamma}}{D\mu^B} dt\rho_0 c_s^2 \delta_{\alpha\beta} = -\int v_\alpha v_\beta \left(\widetilde{f} - f^{eq}\right) d\boldsymbol{v} - \frac{dt}{2}\left(u_\alpha F_\beta + u_\beta F_\alpha\right). \tag{A.12}$$

By setting $\alpha = \beta = \gamma$, formulation for $\sigma_{\gamma\gamma}$ can be obtained, which is subsequently substituted into Eq. (A.12). By incorporating the free parameters in kinematic shear viscosity and bulk viscosity in Eq. (27), finally, the formulation for the viscous stress is given by

$$\sigma_{\alpha\beta} = \frac{-\int v_\alpha v_\beta \left(\widetilde{f} - f^{eq}\right) d\boldsymbol{v} - \frac{dt}{2}\left(u_\alpha F_\beta + u_\beta F_\alpha\right) + M_{\alpha\beta}}{1 + 1/\left[(2\tau_{fL} - 1)(1 - \chi_s)\right]}, \tag{A.13a}$$

where

$$M_{\alpha\beta} = \frac{(\chi_b - \chi_s)\,\delta_{\alpha\beta}\left[\int v_\gamma v_\gamma \left(\widetilde{f} - f^{eq}\right) d\boldsymbol{v} + dt u_\gamma F_\gamma\right]}{D\left[1 - \chi_s + \left(2\tau_{fL} - 1\right)(1 - \chi_s)(1 - \chi_b)\right]}. \tag{A.13b}$$

Appendix A.2. Scalar field

A similar transformation is used to remove the implicit nature of time for the distribution function in the scalar field, as

$$\widetilde{g}(\boldsymbol{v},\boldsymbol{x},t) = g(\boldsymbol{v},\boldsymbol{x},t) + \frac{dt}{2\tau_g}\left[g(\boldsymbol{v},\boldsymbol{x},t) - g^{eq}(\boldsymbol{v},\boldsymbol{x},t)\right] - \frac{dt}{2}\mathcal{R}(\boldsymbol{v},\boldsymbol{x},t). \tag{A.14}$$

Then the explicit LBE is obtained as

$$\widetilde{g}(\boldsymbol{v},\boldsymbol{x}+\boldsymbol{v}dt,t+dt) - \widetilde{g}(\boldsymbol{v},\boldsymbol{x},t) = -\frac{1}{\tau_{gL}}\left[\widetilde{g}(\boldsymbol{v},\boldsymbol{x},t) - g^{eq}(\boldsymbol{v},\boldsymbol{x},t)\right] + \Psi(\boldsymbol{v},\boldsymbol{x},t)dt, \tag{A.15}$$

where $\tau_{gL} = \tau_g/dt + 0.5$, and the source term $\Psi(\boldsymbol{v},\boldsymbol{x},t)$ in LBE is

$$\Psi(\boldsymbol{v},\boldsymbol{x},t) = \left(1 - \frac{1}{2\tau_{gL}}\right)\mathcal{R}(\boldsymbol{v},\boldsymbol{x},t). \tag{A.16}$$

The macroscopic variables, i.e., T. Tu_α are associated with the moments of g (i.e., $g(\boldsymbol{v},\boldsymbol{x},t)$). To obtain them from the LBE, the inverse formulation of Eq. (A.14) is utilized, which is expressed as

$$g(\boldsymbol{v},\boldsymbol{x},t) = \left(1 - \frac{1}{2\tau_{gL}}\right)\widetilde{g}(\boldsymbol{v},\boldsymbol{x},t) + \frac{1}{2\tau_{gL}}g^{eq}(\boldsymbol{v},\boldsymbol{x},t) + \left(1 - \frac{1}{2\tau_{gL}}\right)\frac{dt}{2}\mathcal{R}(\boldsymbol{v},\boldsymbol{x},t). \tag{A.17}$$

Therefore, these variables can be derived from the moments of $\widetilde{g}$, as

$$T \equiv \int g d\boldsymbol{v} = \left(1 - \frac{1}{2\tau_{gL}}\right)\int \widetilde{g} d\boldsymbol{v} + \frac{T}{2\tau_{gL}} + \left(1 - \frac{1}{2\tau_{gL}}\right)\frac{dt}{2}Q, \tag{A.18}$$

which can be rewritten and simplified as

$$T = \int g d\boldsymbol{v} = \int \widetilde{g} d\boldsymbol{v} + \frac{dt}{2}Q. \tag{A.19}$$

Additionally, the heat flux is expressed as

$$\begin{aligned} q = -\kappa\partial_\alpha T &= \int v_\alpha(g - g^{eq})d\boldsymbol{v} \\ &= \frac{2\int v_\alpha\left(\widetilde{g} - g^{eq}\right)d\boldsymbol{v} + dt\left(TF_\alpha/\rho_0 + u_\alpha Q\right)}{2 + dt(p/\rho_0 + c_s^2)/\kappa}. \end{aligned} \tag{A.20}$$

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